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Jeong et al.

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(54) **METHOD AND APPARATUS FOR
ENCODING AND DECODING OF LOW
DENSITY PARITY CHECK CODES**

(58) **Field of Classification Search**
CPC H03M 13/1165; H03M 13/255; H03M
13/2906; H03M 13/6552; H03M 13/036;
(Continued)

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U.S.C. 154(b) by 20 days.

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This patent is subject to a terminal dis-
claimer.

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(57) **ABSTRACT**

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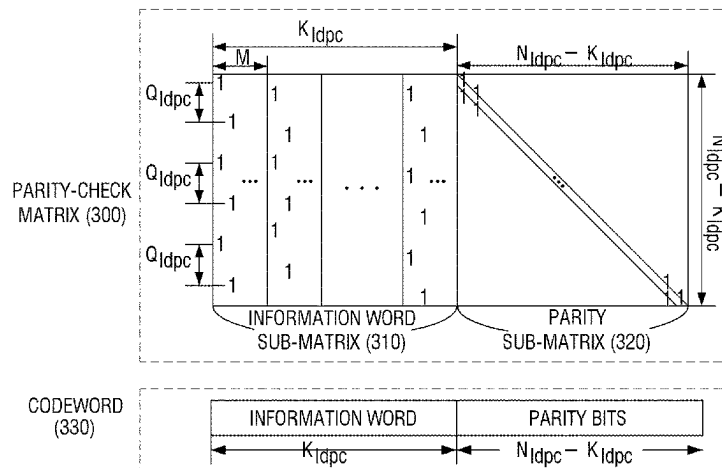
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CPC **H03M 13/1165** (2013.01); **H03M 13/255**
(2013.01); **H03M 13/2906** (2013.01);

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An encoding apparatus is provided. The encoding includes
a low density parity check (LDPC) encoder which performs
LDPC encoding on input bits based on a parity-check matrix
to generate an LDPC codeword formed of 64,800 bits, in
which the parity-check matrix includes an information word
sub-matrix and a parity sub-matrix, the information word
sub-matrix is formed of a group of a plurality of column
blocks each including 360 columns, and the parity-check
matrix and the information word sub-matrix are defined by
various tables which represent positions of value one (1)
present in every 360-th column.

8 Claims, 14 Drawing Sheets



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CPC H03M 13/152; H03M 13/1102; H03M 13/015; H03M 13/27; H04L 1/0011

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See application file for complete search history.

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FIG. 1

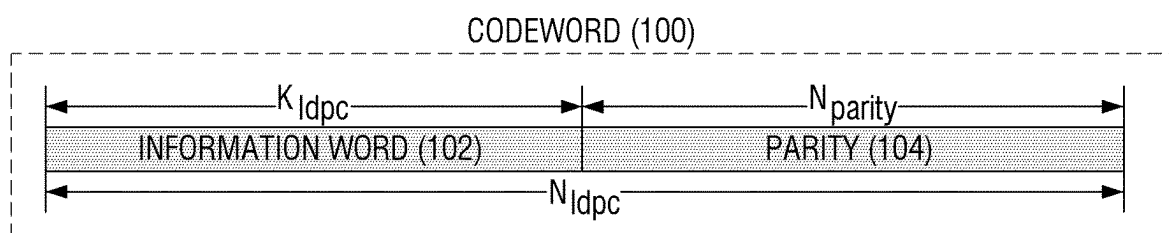


FIG. 2

$$H_1 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

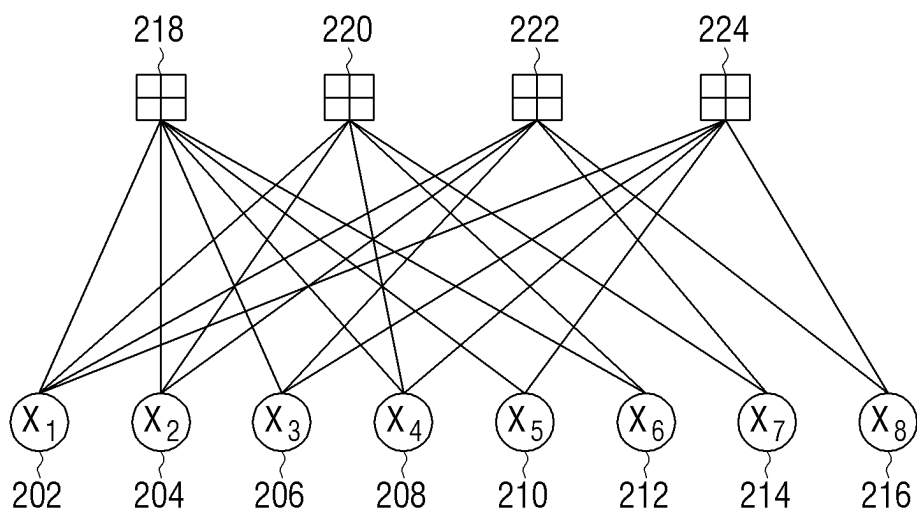


FIG. 3

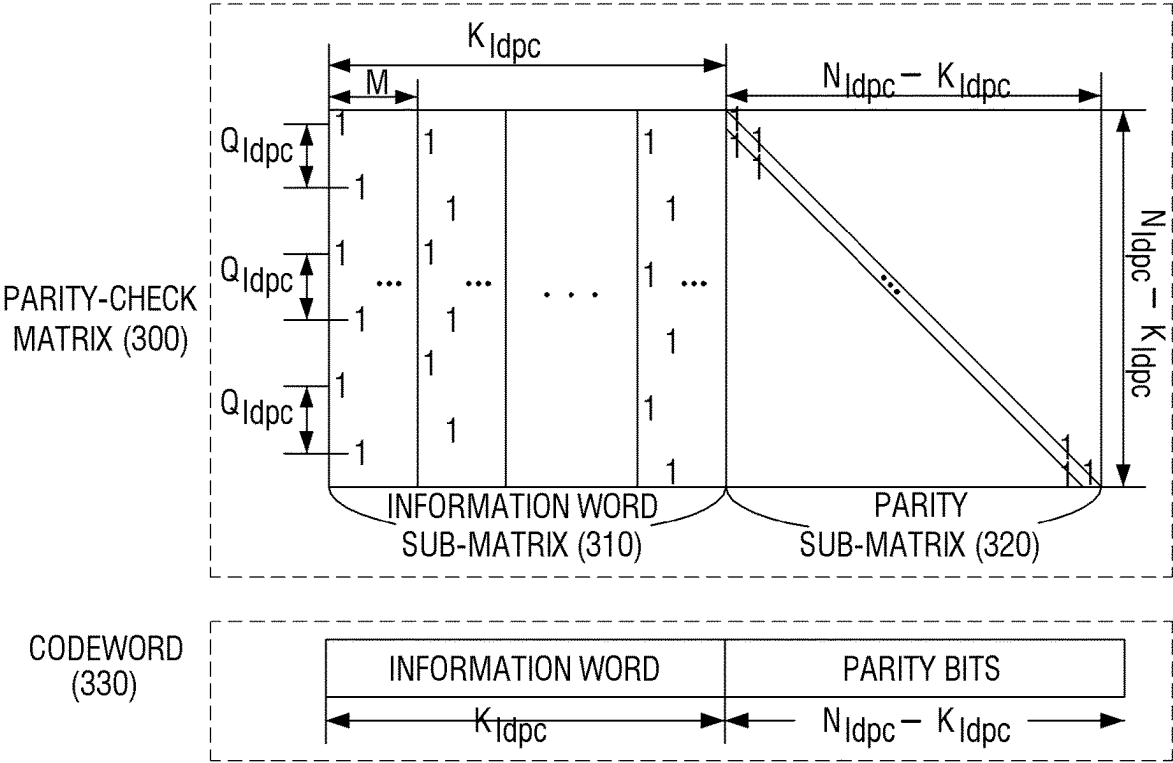


FIG. 4

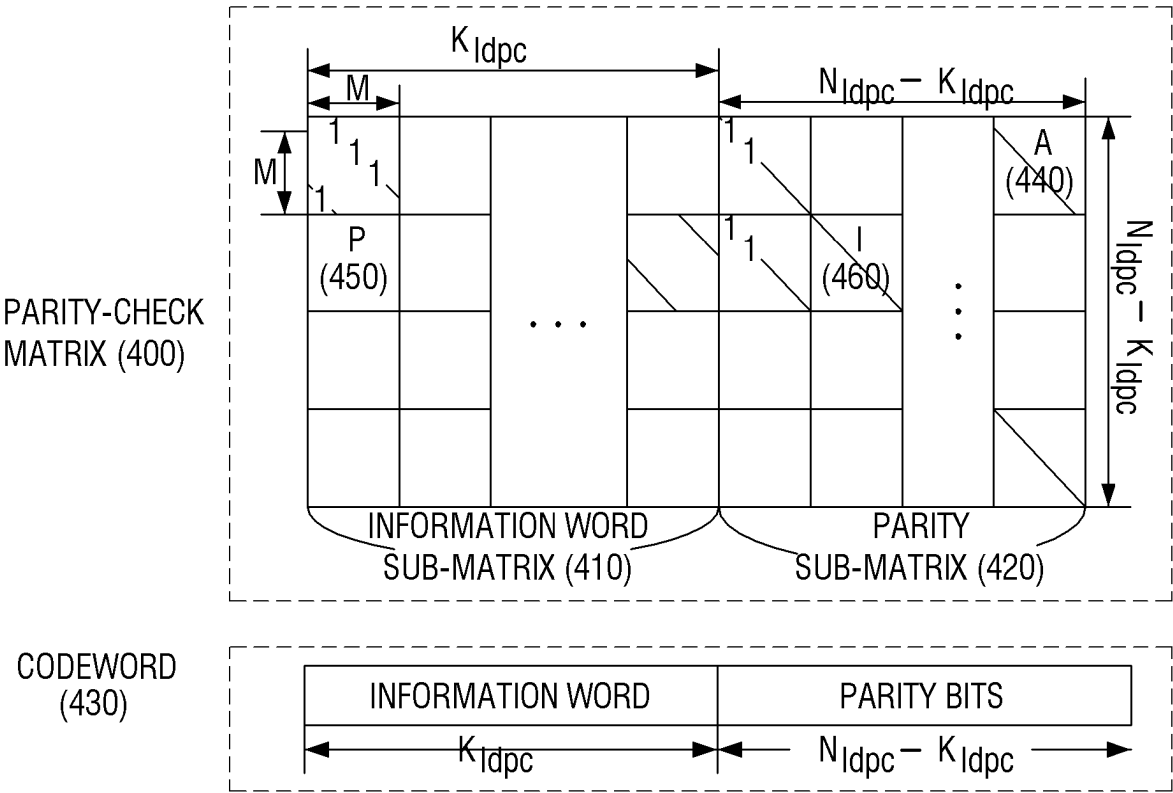


FIG. 5A

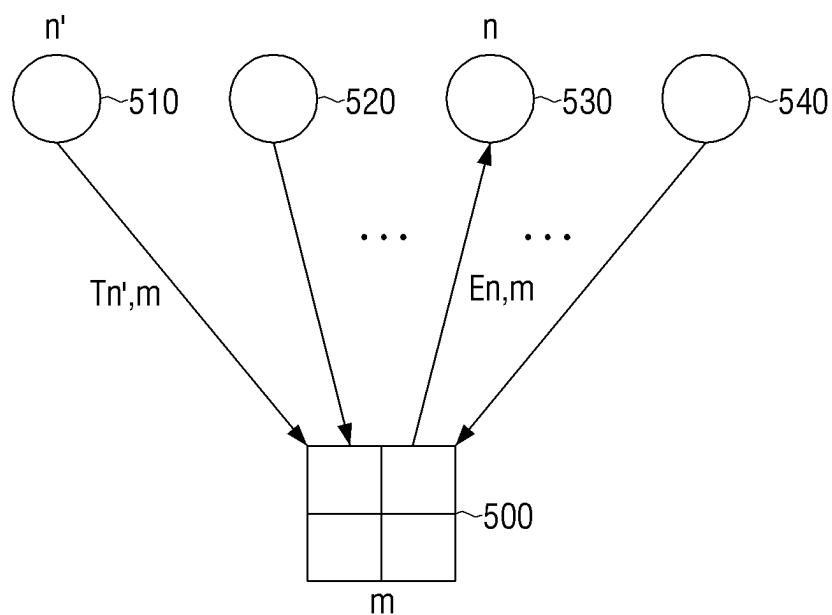


FIG. 5B

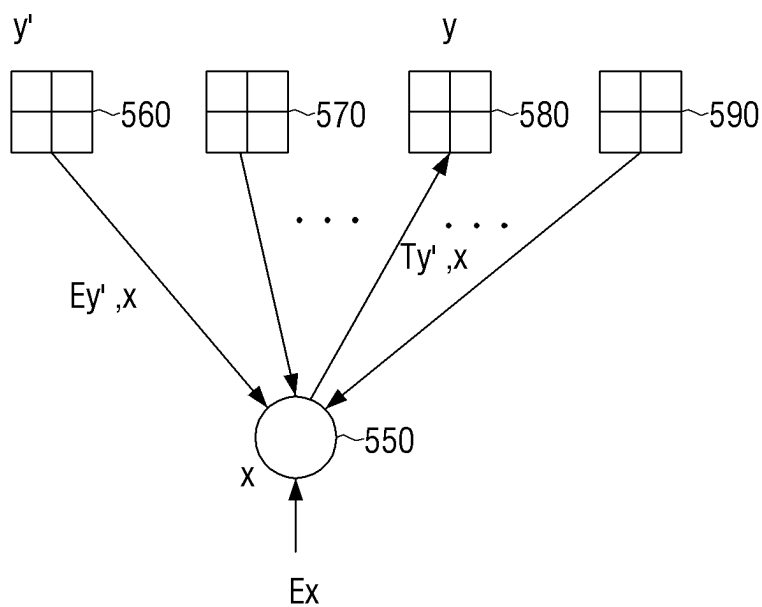


FIG. 6

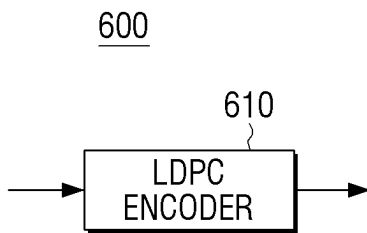


FIG. 7

700

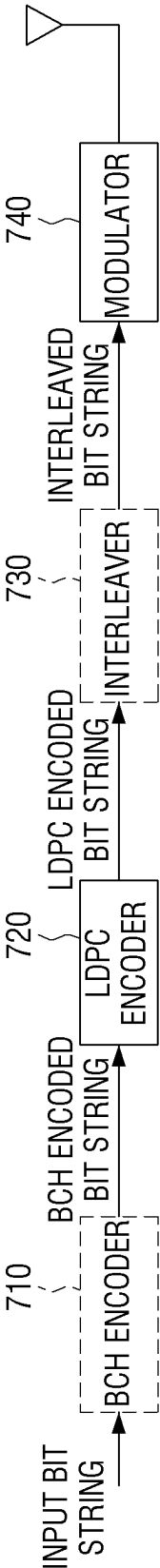


FIG. 8

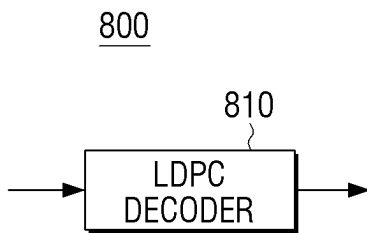


FIG. 9

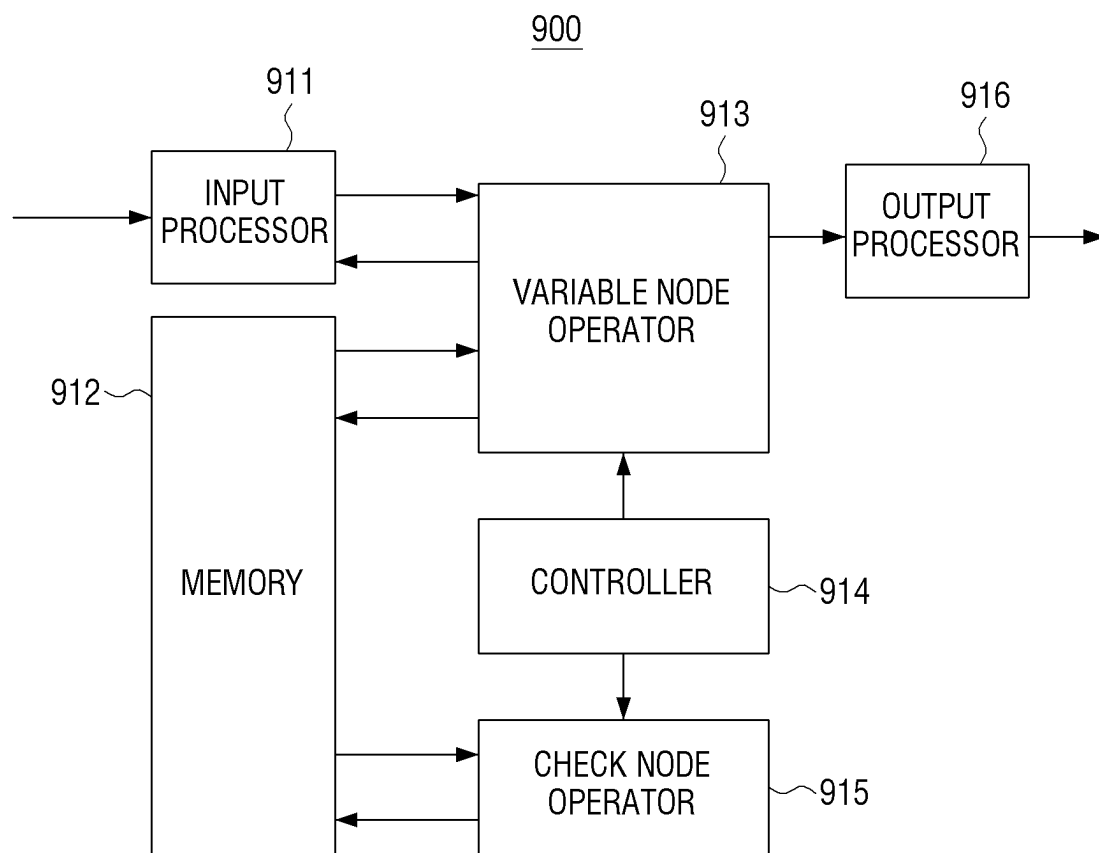


FIG. 10

1000

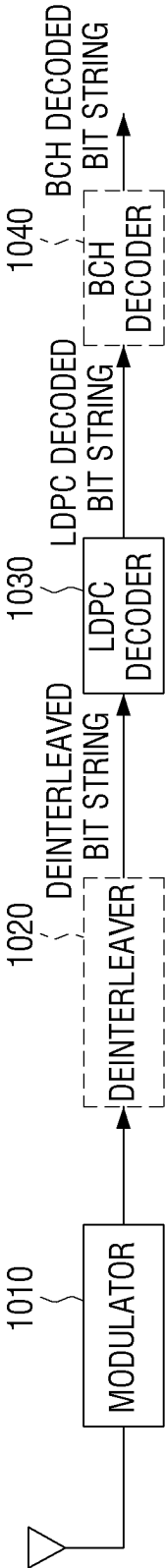


FIG. 13

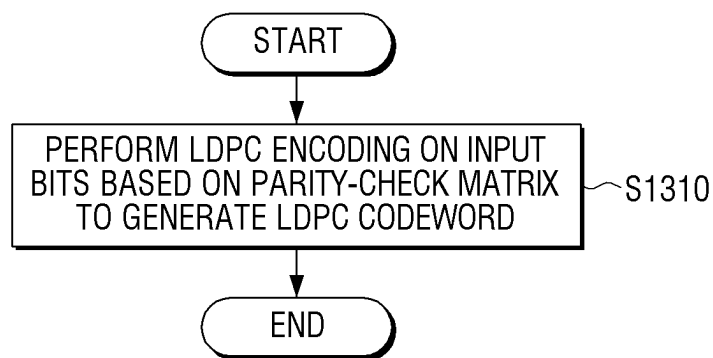
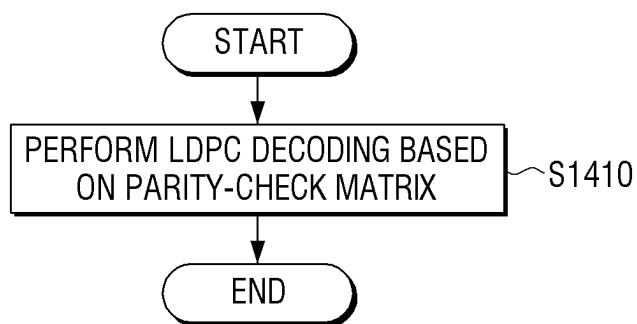


FIG. 14



1

METHOD AND APPARATUS FOR ENCODING AND DECODING OF LOW DENSITY PARITY CHECK CODES

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation of U.S. application Ser. No. 17/688,348 filed Mar. 7, 2022, which is a continuation of U.S. application Ser. No. 16/523,369 filed Jul. 26, 2019, which is a continuation of U.S. application Ser. No. 14/303,834 filed Jun. 13, 2014 which is now U.S. Pat. No. 10,411,736 issued Sep. 10, 2019, which claims priority from Korean Patent Application Nos. 10-2014-0058599, filed on May 15, 2014, in the Korean Intellectual Property Office, and U.S. Provisional Application Nos. 61/862,208, 61/841,502, and 61/835,096, filed on Aug. 5, 2013, Jul. 1, 2013, and Jun. 14, 2013, respectively, in the United States Patent and Trademark Office, the disclosures of which are incorporated herein by reference in their entirety.

BACKGROUND

1. Field

Apparatuses and methods consistent with the exemplary embodiments of the inventive concept relate to encoding and decoding Low Density Parity Check (LDPC) codes, and more particularly, to encoding and decoding LDPC codes which perform LDPC encoding and decoding based on a parity-check matrix.

2. Description of the Related Art

In a communication/broadcasting system, link performance may considerably deteriorate due to various types of noises, a fading phenomenon, and inter-symbol interference (ISI) of a channel. Therefore, to implement high-speed digital communication/broadcasting systems requiring high data throughput and reliability like next-generation mobile communications, digital broadcasting, and portable Internet, it has been required to develop technologies to overcome the noises, the fading, and the inter-symbol interference. As part of studies to overcome the noises, etc., a study on an error-correcting code which is a method for increasing reliability of communications by efficiently recovering distorted information has been actively conducted recently.

LDPC codes which were first introduced by Gallager in the 1960s remain forgotten for a very long time due to complexity which could hardly be implemented at the technology level at that time. However, as performance of turbo codes proposed by Berrou, Glavieux, and Thitimajshima in 1993 approaches Shannon's channel capacity, many studies on channel encoding based on iterative decoding and a graph thereof by performing many different interpretations on performance and characteristics of the turbo codes have been conducted. As a result, as the LDPC codes in the late 1990s are restudied, when the LDPC codes are decoded by applying sum-product algorithm based iterative decoding to the LDPC codes on a tanner graph corresponding to the LDPC codes, it was found that the performance of the LDPC codes approaches the Shannon's channel capacity.

The LDPC codes may be generally defined by a parity-check matrix and represented using a bipartite graph commonly referred to as the tanner graph.

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Hereinafter, a systematic LDPC codeword will be described with reference to FIG. 1. The LDPC codes are LDPC encoded by receiving an information word **102** formed of K_{ldpc} bits or symbols to generate a codeword **100** formed of N_{ldpc} bits or symbols. Hereinafter, for convenience of explanation, it is assumed that the codeword **100** formed of N_{ldpc} bits is generated by receiving the information word **102** including K_{ldpc} bits. That is, when the information word $l=[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}]$ **102** which is formed of K_{ldpc} input bits is LDPC encoded, the codeword $c=[c_0, c_1, c_2, c_3, \dots, c_{N_{ldpc}-1}]$ **100** is generated. That is, the codeword is a bit string formed of a plurality of bits and the bits of the codeword represent each bit forming the codeword. Further, the information word is a bit string formed of a plurality of bits and the bits of the information word represent each bit forming the information word. In this case, in the case of a systematic code, the codeword is formed of $c=[c_0, c_1, c_2, \dots, c_{N_{ldpc}-1}]=[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}, p_0, p_1, p_2, \dots, p_{N_{ldpc}-K_{ldpc}-1}]$. Here, $P=[p_0, p_1, p_2, \dots, p_{N_{ldpc}-K_{ldpc}-1}]$ is a parity **104** and the number N_{parity} of parity bits is as follows. $N_{parity}=N_{ldpc}-K_{ldpc}$.

The LDPC codes are a kind of linear block codes and include a process of determining a codeword satisfying conditions of following mathematical expression 1.

$$H \cdot c^T = [h_0, h_1, h_2, \dots, h_{N_{ldpc}-1}] \cdot c^T = \sum_{i=0}^{N_{ldpc}-1} c_i \cdot h_i = 0 \quad (1),$$

where $c=[c_0, c_1, c_2, \dots, c_{N_{ldpc}-1}]$.

In mathematical expression 1 above, H represents the parity-check matrix, C represents the codeword, c_i represents an i-th bit of the codeword, and N_{ldpc} represents a codeword length. Here, h_i represents an i-th column of the parity-check matrix H.

The parity-check matrix H is formed of the same N_{ldpc} columns as the number of bits of the LDPC codeword. The mathematical expression 1 represents that since a sum of a product of the i-th column h_i of the parity-check matrix and the i-th bit c_i of the codeword becomes "0", the i-th column h_i has a relationship with the i-th bit c_i of the codeword.

Meanwhile, the performance of the LDPC codes may be determined according to the parity-check matrix. Therefore, there is a need to design the parity-check matrix for the LDPC codes having improved performance.

SUMMARY

One or more exemplary embodiments may overcome the above disadvantages and other disadvantages not described above. However, these embodiments are not required to overcome the disadvantages described above and may not overcome any of the problems described above.

One or more exemplary embodiments provide a method and an apparatus for encoding and decoding LDPC codes capable of improving LDPC encoding and decoding performance.

According to an aspect of an exemplary embodiment, there is provided an encoding apparatus which may include: an LDPC encoder configured to perform LDPC encoding on input bits based on a first parity-check matrix or a second parity-check matrix to generate an LDPC codeword formed of 64,800 bits, wherein the parity-check matrix includes an information word sub-matrix and a parity sub-matrix, and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.

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The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 17 when a code rate is 6/15.

The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

The LDPC encoder may perform the LDPC encoding using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

According to an aspect of another exemplary embodiment, there is provided an encoding method which may include: generating an LDPC codeword formed of 64,800 bits by performing LDPC encoding on input bits based on a parity-check matrix, in which the parity-check matrix includes an information word sub-matrix and a parity sub-matrix and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.

In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 17 when a code rate is 6/15.

In the generating of the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

In the generating the LDPC codeword, the LDPC encoding may be performed using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

According to an aspect of still another exemplary embodiment, there is provided a decoding apparatus which may include: an LDPC decoder configured to perform LDPC decoding on an LDPC codeword formed of 64,800 bits based on a parity-check matrix, wherein the parity-check matrix includes an information word sub-matrix and a parity sub-matrix, and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) presented in every 360-th column.

The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 17 when the code rate is 6/15.

The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

The LDPC decoder may perform the LDPC decoding using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

According to an aspect of still another exemplary embodiment, there is provided a decoding method which may include: performing LDPC decoding on an LDPC codeword formed of 64,800 bits based on a parity-check matrix, in which the parity-check matrix includes an information word sub-matrix and a parity sub-matrix and the information word sub-matrix is formed of a plurality of column blocks each including 360 columns, and is defined by a table which represents positions of value one (1) present in every 360-th column.

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In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 17 when the code rate is 6/15.

In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 14 when the code rate is 8/15.

In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 18 when the code rate is 10/15.

In the performing the LDPC decoding, the LDPC decoding may be performed using a parity-check matrix defined by a table like Table 16 when the code rate is 12/15.

Additional and/or other aspects and advantages of the exemplary embodiments will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of these embodiments.

BRIEF DESCRIPTION OF THE DRAWING FIGURES

The above and/or other aspects will be more apparent by describing certain exemplary embodiments with reference to the accompanying drawings, in which:

FIG. 1 is a diagram illustrating a codeword of a systematic LDPC code;

FIG. 2 is a diagram illustrating a parity-check matrix and a factor graph of general (8, 2, 4) LDPC codes.

FIG. 3 is a diagram illustrating a parity-check matrix according to an exemplary embodiment;

FIG. 4 is a diagram illustrating the parity-check matrix generated by permuting a row and a column of the parity-check matrix of FIG. 3, according to an exemplary embodiment;

FIGS. 5A and 5B are diagrams illustrating a check node and a variable node used for LDPC decoding, according to exemplary embodiments;

FIG. 6 is a block diagram for describing a configuration of an encoding apparatus, according to an exemplary embodiment;

FIG. 7 is a block diagram for describing a configuration of a transmitting apparatus, according to an exemplary embodiment;

FIG. 8 is a block diagram for describing a configuration of a decoding apparatus, according to an exemplary embodiment;

FIG. 9 is a block diagram for describing a decoding apparatus, according to an exemplary embodiment;

FIG. 10 is a block diagram for describing a configuration of a receiving apparatus, according to an exemplary embodiment;

FIGS. 11 and 12 are diagrams for describing an effect generated at the time of performing LDPC encoding, according to exemplary embodiments;

FIG. 13 is a flow chart for describing an encoding method, according to an exemplary embodiment; and

FIG. 14 is a flow chart for describing a decoding method, according to an exemplary embodiment.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

Hereinafter, exemplary embodiments will be described in detail with reference to the accompanying drawings. Further, in describing the exemplary embodiments, detailed

descriptions of well-known functions or constructions will be omitted so as not to obscure the description with unnecessary detail.

Hereinafter, the exemplary embodiments will describe a technology of LDPC encoding and LDPC decoding in a communication/broadcasting system.

Hereinafter, the exemplary embodiments use terms and names which are defined in the Digital Video Broadcasting the Second Generation Terrestrial (DVB-T2) system which is one of the European digital broadcasting standards and North America digital broadcasting standard system, Advanced Television Systems Committee (ATSC) 3.0 which is being established as standard. However, the inventive concept is not limited to these terms and names, but may be similarly applied to other systems.

A graph representation method of LDPC codes will be described with reference to FIG. 2.

FIG. 2 is a diagram illustrating an example of a parity-check matrix H_1 of the LDPC codes which is formed of four (4) rows and eight (8) columns. The parity-check matrix H_1 is represented with a tanner graph. Referring to FIG. 2, the parity-check matrix H_1 has eight (8) columns, and thus, a codeword having a length of eight (8) is generated. The Codes generated through the H_1 represent the LDPC codes, and each column of the H_1 corresponds to encoded eight (8) bits.

Referring to FIG. 2, the tanner graph of the LDPC codes which are encoded and decoded based on the parity-check matrix H_1 is formed of eight (8) variable nodes, that is, x_1 202, x_2 204, x_3 206, x_4 208, x_5 210, x_6 212, x_7 214 and x_8 216 and four check nodes 218, 220, 222 and 224. Here, an i -th column and a j -th row of the parity-check matrix H_1 of the LDPC codes each correspond to a variable node x_i and a j -th check node, respectively. Further, a value of one (1) at an intersection point of a j -th column and a j -th row of the parity-check matrix H_1 of the LDPC codes, that is, a non-zero value represents that an edge connecting between the variable node x_i and the j -th check node on the tanner graph as illustrated in FIG. 2 exists.

A degree of the variable node and the check node on the tanner graph of the LDPC codes represents the number of edges connected to each node, which is the same as the number of non-zero entries in a row or a column corresponding to the corresponding node in the parity-check matrix of the LDPC codes. For example, in FIG. 2, the degrees of the variable nodes x_1 202, x_2 204, x_3 206, x_4 208, x_5 210, x_6 212, x_7 214 and x_8 216 each become 4, 3, 3, 3, 2, 2, 2 and 2 in order, and the degrees of the check nodes 218, 220, 222 and 224 each become 6, 5, 5 and 5 in order. Further, the number of non-zero entries in each column of the parity-check matrix H_1 of FIG. 2 corresponding to the variable nodes of FIG. 2 coincides with the degrees 4, 3, 3, 3, 2, 2, 2 and 2 in order, and in each row of the parity-check matrix H_1 of FIG. 2 corresponding to the check nodes of FIG. 2, the number of non-zero entries coincides with the degrees 6, 5, 5 and 5 in order.

The LDPC codes may be decoded using an iterative decoding algorithm based on a sum-product algorithm on a bipartite graph illustrated in FIG. 2. Herein, the sum-product algorithm is a kind of message passing algorithm, and the message passing algorithm represents an algorithm which exchanges messages through the edge on the bipartite graph and calculates output messages using messages input to the variable nodes or the check nodes to update the messages.

Herein, a value of an i -th encoding bit may be determined based on a message of an i -th variable node. A soft decision and a hard decision may be performed on the value of the

i -th encoding bit. Therefore, performance of an i -th bit c_i of the LDPC codeword corresponds to performance of the i -th variable node, which may be determined depending on positions and the number of ones (1s) in the i -th column of the parity-check matrix. That is, performance of N_{ldpc} bits of the codeword relies on the positions and number of ones (1s) in the parity-check matrix.

Hereinafter, characteristics of the parity-check matrix of the LDPC codes having a specific structure will be described with reference to FIG. 3.

FIG. 3 illustrates a parity-check matrix having a structure according to an exemplary embodiment. The parity-check matrix illustrated in FIG. 3 has a structure which may encode and decode a systematic code in which the codeword includes an original information word. Hereinafter, according to an exemplary embodiment will be described based on the parity-check matrix of FIG. 3, but a scope of application of the inventive concept is not limited to the parity-check matrix as illustrated in FIG. 3.

In FIG. 3, N_{ldpc} represents the LDPC codeword length and K_{ldpc} represents the information word length. Meanwhile, the codeword length or the information word length represents the number of bits included in the codeword or the information word, respectively. M represents an interval at which a pattern of a column is repeated in a sub-matrix 310 (hereinafter, referred to as an information word sub-matrix) corresponding to the information word and Q_{ldpc} is a size of cyclic shift of each column in the information word sub-matrix 310, in which values of the integer M and the Q_{ldpc} are determined to be $Q_{ldpc} = (N_{ldpc} - K_{ldpc}) / M$. Here, K_{ldpc} / M is also an integer. Values of the M and the Q_{ldpc} may be changed depending on the codeword length and a code rate (or coding rate).

Referring to FIG. 3, the parity-check matrix 300 is divided into the information word sub-matrix 310 (or information word matrix) corresponding to the information word and a parity sub-matrix 320 (or parity matrix) corresponding to the parity. The information word sub-matrix 310 includes K_{ldpc} columns and the parity sub-matrix 320 includes $N_{parity} = N_{ldpc} - K_{ldpc}$ columns. The number of rows of the parity-check matrix 300 and the number of columns of the parity sub-matrix 320 are equal to $N_{ldpc} - K_{ldpc}$.

Positions of entries having weight-1s, that is, value one (1) present in the parity sub-matrix 320 including a K_{ldpc} -th column to an $(N_{ldpc} - 1)$ -th column of the parity-check matrix 300 have a dual diagonal structure. Therefore, among the columns included in the parity sub-matrix 320, all the degrees (herein, the degree is the number of ones (1s) included in each column) of the remaining columns except the $(N_{ldpc} - 1)$ -th column are two (2) and the degree of the $(N_{ldpc} - 1)$ -th column is one (1).

Meanwhile, a structure of the information word sub-matrix 310, that is, the sub-matrix including a zero (0)-th column to a $(K_{ldpc} - 1)$ -th column depends on the following rule.

First, the K_{ldpc} columns corresponding to the information word in the parity-check matrix 300 belong to a plurality of column groups each having M columns, and the information word sub-matrix 310 is divided into a total of K_{ldpc} / M column groups. In columns belonging to the same column group, positions of rows at which one (1) is present have a relationship in which a row having a value of one (1) is shifted by Q_{ldpc} from an immediately previous row having the same value of one (1). Here, Q_{ldpc} may be an integer greater than or equal to one (1).

Second, if it is assumed that a degree of a zero (0)-th column of an i -th ($i=0, 1, \dots, K_{ldpc}/M-1$) column group is

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D_i and positions of each row at which one (1) is positioned are $R_{i,0}^{(0)}, R_{i,0}^{(1)}, \dots, R_{i,0}^{(D_i-1)}$, an index $R_{i,j}^{(k)}$ of a row at which a k -th weight-1, i.e., k -th one (1), is positioned in a j -th column of the i -th column group is determined as represented by following mathematical expression 2.

$$R_{i,j}^{(k)} = (R_{i,(i-1)}^{(k)} + Q_{ldpc}) \bmod (N_{ldpc} - K_{ldpc}) \quad (2)$$

In mathematical expression 2 above, $k=0, 1, 2, \dots, D_i-1$, $i=0, 1, \dots, K_{ldpc}/M-1$, and $j=1, 2, \dots, M-1$.

The above mathematical expression 2 may be represented like following mathematical expression 3.

$$R_{i,j}^{(k)} = (R_{i,0}^{(k)} + (j \bmod M) \times Q_{ldpc}) \bmod (N_{ldpc} - K_{ldpc}) \quad (3)$$

In mathematical expression 3 above, $k=0, 1, 2, \dots, D_i-1$, $i=0, 1, \dots, K_{ldpc}/M-1$, and $j=1, 2, \dots, M-1$.

In the above mathematical expressions, $R_{i,j}^{(k)}$ represent the indices of the row at which the k -th weight-1 is positioned in the j -th column of the i -th column group, N_{ldpc} represents the LDPC codeword length, K_{ldpc} represents the information word length, D_i represents the degree of a zero (0)-th column belonging to the i -th column group, and M represents the number of columns belonging to one column group.

According to the above mathematical expressions, when only a value of $R_{i,0}^{(k)}$ is known, the indices of the row at which the k -th weight-1 is positioned in the i -th column group may be known. Therefore, when index values of the row at which the k -th weight-1 is positioned in the zero (0)-th column of each column group are stored, the positions of the column and the row at which the weight-1 is positioned may be understood in the parity-check matrix **300** (that is, the information word sub-matrix **310** of the parity-check matrix **300**) having the structure illustrated in FIG. 3.

According to the above-mentioned rules, all of the degrees of the columns belonging to the i -th column group are D_i . According to the above-mentioned rules, the LDPC codes in which information on the parity-check matrix **300** is stored may be simply represented as follows.

As a specific example, when N_{ldpc} is 30, K_{ldpc} is 15, and Q_{ldpc} is three (3), position information of rows at which the weight-1 is positioned in zero (0)-th columns of three column groups may be represented by sequences as represented by following mathematical expression 4. The sequences represented by mathematical expression 4 may be called a weight-1 position sequence.

$$\begin{aligned} R_{1,0}^{(1)} &= 1, R_{1,0}^{(2)} = 2, R_{1,0}^{(3)} = 8, R_{1,0}^{(4)} = 10, \\ R_{2,0}^{(1)} &= 0, R_{2,0}^{(2)} = 9, R_{2,0}^{(3)} = 13, \\ R_{3,0}^{(1)} &= 0, R_{3,0}^{(2)} = 14. \end{aligned} \quad (4)$$

In mathematical expression 4 above, $R_{i,j}^{(k)}$ represents the indices of the row at which the k -th weight-1 is positioned in the j -th column of the i -th column group.

The weight-1 position sequences as represented by the above mathematical expression 4, which represents the indices of the rows at which one (1) is positioned in the zero (0)-th column of each column group, may be more simply represented like following Table 1.

TABLE 1

1 2 8 10
0 9 13
0 14

Table 1 above shows the positions of entries having the weight-1, that is, value one (1) in the parity-check matrix, in

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which an i -th weight-1 position sequence is represented by the indices of the row at which the weight-1 is positioned in the zero (0)-th column belonging to the i -th column group.

When the column and the row of the parity-check matrix **300** illustrated in FIG. 3 are permuted by following mathematical expression 5 (row permutation) and mathematical expression 6 (column permutation), the parity-check matrix **300** illustrated in FIG. 3 may be shown as a form of a parity-check matrix **400** illustrated in FIG. 4.

$$Q_{ldpc} \cdot i + j \Rightarrow M \cdot j + i \quad (0 \leq i < M, 0 \leq j < Q_{ldpc}) \quad (5)$$

$$i \Rightarrow i \quad (0 \leq i < K_{ldpc})$$

$$K_{ldpc} + Q_{ldpc} \cdot k + l \Rightarrow K_{ldpc} + M \cdot l + k \quad (0 \leq k < M, 0 \leq l < Q_{ldpc}) \quad (6)$$

In the above, the row permutation represents that an order of the rows of the parity-check matrix **300** is changed using mathematical expression 5 above. Further, the column permutation represents that an order of columns of the parity-check matrix **300** is changed using mathematical expression 6 above.

A method of performing permutation based on mathematical expressions 5 and 6 above is as follows. In this case, the column permutation is applied with the same principle as the row permutation, except the fact that the column permutation is applied only to the parity sub-matrix **320** by $i \Rightarrow i \quad (0 \leq i < K_{ldpc})$. Hereinafter, the row permutation will be described as an example.

In the case of the row permutation, i and j meeting $X = Q_{ldpc} \times i + j$ for an X -th row are calculated, and the calculated i and j are substituted in $M \times j + i$ to calculate a row in which the X -th row is permuted. For example, in the case of a seventh row, i and j meeting $7 = 2 \times i + j$ each are 3 and 1 and therefore the seventh row is permuted to a thirteenth ($10 \times 1 + 3 = 13$) row.

When the row and column of the parity-check matrix of FIG. 3 are each permuted depending on the mathematical expressions 5 and 6, the parity-check matrix may be shown in a form of the parity-check matrix illustrated in FIG. 4.

Referring to FIG. 4, the parity-check matrix **400** of the LDPC codes has a form in which the entire parity-check matrix **400** is divided into a plurality of partial blocks and each of the partial blocks corresponds to an $M \times M$ quasi-cyclic matrix.

The parity-check matrix **400** illustrated in FIG. 4 is formed in a unit of the $M \times M$ quasi-cyclic matrix, and thus, M columns may be called a column-block (or column group) and M rows may be called a row-block (or row group). That is, the parity-check matrix **400** having the form of FIG. 4 which is used in the present exemplary embodiment is formed of $N_{qc_column} = N_{ldpc}/M$ column-blocks and $N_{qc_row} = N_{parity}/M$ row-blocks.

Hereinafter, $M \times M$ matrices forming the parity-check matrix **400** having the form of FIG. 4 will be described in detail.

First, an $M \times M$ matrix **440** of an $(N_{qc_column} - 1)$ -th column-block among the zero (0)-th row-blocks has a form as represented by following mathematical expression 7.

$$A = \begin{bmatrix} 0 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix} \quad (7)$$

In the $M \times M$ matrix as represented by mathematical expression 7 above, all the values of the zero (0)-th row and an $M-1$ -th column are '0'. For $0 \leq i \leq (M-2)$, an $(i+1)$ -th row of the i -th column is '1' and all the other values are '0'.

Second, in the parity sub-matrix **420** of the parity-check matrix **400** of FIG. 4, for $0 \leq i \leq (N_{ldpc} - K_{ldpc})/M - 1$, an i -th row-block of an $(K_{ldpc}/M + i)$ -th column-blocks is formed of unit matrices $I_{M \times M}$ **460**. Further, for $0 \leq i \leq (N_{ldpc} - K_{ldpc})/M - 2$, an $(i+1)$ -th row block of the $K_{ldpc}/M + i$ -th column-blocks is formed of the unit matrices $I_{M \times M}$ **460**.

Third, the information word sub-matrix **410** may be a form in which a matrix $P^{a_{ij}}$ obtained by cyclically shifting a quasi-cyclic matrix P , or the quasi-cyclic matrix P and a matrix $P^{a_{ij}}$ obtained by cyclically shifting the quasi-cyclic matrix P are combined.

The quasi-cyclic matrix P is represented by following mathematical expression 8.

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (8)$$

The quasi-cyclic matrix P of the above mathematical expression 8 is a square matrix having an $M \times M$ size and is a matrix of which the respective degrees of M rows and columns forming the quasi-cyclic matrix P are one (1).

If a subscript a_{ij} of the quasi-cyclic matrix P is 0, the quasi-cyclic matrix P^0 represents a unit matrix $I_{M \times M}$, and if the subscript a_{ij} of the quasi-cyclic matrix P is ∞ , the quasi-cyclic matrix P^∞ represents a zero matrix. As illustrated in FIG. 4, in the entire parity-check matrix of the LDPC codes, the total number of columns is $N_{ldpc} = M \times N_{qc_column}$ and the total number of rows is $N_{parity} = M \times N_{qc_row}$. That is, the parity-check matrix **400** as illustrated in FIG. 4 is formed of N_{qc_column} column-blocks and N_{qc_row} row-blocks.

The parity-check matrix **400** as illustrated in FIG. 4 is represented by positions of a non-zero quasi-cycle matrix and index values at the corresponding positions.

In the case of the LDPC codes, the codeword performance is determined according to the parity-check matrix. Specifically, the codeword performance may be determined according to a weight distribution and a cycle distribution of columns and rows.

The weight distribution of columns represents how many columns have one (1) and how many ones (1s) are positioned in the columns, in the N_{ldpc} columns. Further, the weight distribution of rows represents how many rows have one (1) and how many ones (1s) are positioned in the rows, in the $N_{ldpc} - K_{ldpc}$ rows. Further, a weight or a degree of one (1) represents the number of ones (1s) of each column and row.

Herein, the weight distribution of columns and rows may be determined based on a method called density evolution (reference: Richardson, T., and URBANKE, R.: 'The capacity of low-density parity-check codes under message-passing decoding', IEEE Trans. Inf. Theory, 2001, 47, (2), pp. 599-618).

In detail, in the case of using the density evolution method, when the LDPC encoding/decoding are performed based on the parity-check matrix having the given degree distribution, it may be estimated how many times of iteration is required for coding error probability to be '0' at any signal to noise ratio (SNR). In the case of the density evolution,

since it is estimated whether the coding error probability is '0' under the assumption that the codeword length is infinite, the degree distribution of the parity-check matrix may not be determined only by the density evolution if the parity-check matrix for codes having a finite length is designed.

Further, the number of ones (1s) depending on the degree distribution of the parity-check matrix affects encoding/decoding complexity, and thus, the parity-check matrix needs to be designed based on the code performance which is verified not only based on a theoretical approach called the density evolution but also on the actual encoding/decoding complexity and the designed parity-check matrix.

Hereinafter, a method for designing a parity-check matrix will be described in detail.

The form of a parity-check matrix may be very variously present, but the inventive concept intends to design a parity-check matrix having a specific form illustrated in FIG. 3. Further, a parity-check matrix determines the degree distribution, in which a coding gain may be maximal, based on the density evolution, and by a cycle removing method, an error floor occurs in an area in which BER/FER is low. As described above, the parity-check matrix **300** having the form of FIG. 3 turns into the parity-check matrix **400** having the form of FIG. 4 by permuting the columns and the rows.

In the case of the form of FIG. 4, the parity-check matrix may be designed based on the $M \times M$ matrix, and thus, may be easily designed. Therefore, according to an exemplary embodiment, a parity-check matrix is designed in the form of the parity-check matrix **400** having the form of FIG. 4, and then, the parity-check matrix **300** having the form of FIG. 3 is designed by permuting the rows and the columns.

Hereinafter, the method for designing a parity-check matrix according to an exemplary embodiment will be described in detail.

Step 1) The sizes N_{ldpc} and K_{ldpc} and the values of M and Q_{ldpc} of the parity-check matrix to be designed are determined.

According to an exemplary embodiment, the foregoing parameters of the parity-check matrix may be determined as following Table 2.

TABLE 2

code rate	N_{ldpc}	K_{ldpc}	N_{parity}	M	Q_{ldpc}	N_{qc_column}	N_{qc_row}
$5/15$	64800	25920	38880	360	108	180	108
$8/15$	64800	34560	30240	360	84	180	84
$10/15$	64800	43200	21600	360	60	180	60
$12/15$	64800	51840	12960	360	36	180	36

In the case of the parity-check matrix **400** having the form of FIG. 4, a parity sub-matrix **420** is fixed, and thus, only the positions and distributions of one (1) in the information word sub-matrix **410** need to be determined. Further, the positions and distributions of one (1) in the information word sub-matrix **410** are formed in a unit of $M \times M$ quasi-cyclic matrix, and thus, the number and positions of quasi-cyclic matrix, which is not a zero matrix, and cyclic shift values which are the index values of the quasi-cyclic matrices are determined.

Hereinafter, in step 2, the number of quasi-cyclic matrix, not the zero matrix, is determined.

Step 2) The degree distribution of the parity-check matrix is determined by the density evolution method.

As described above, the distribution of one (1) in the parity-check matrix dominates the performance of the LDPC codes. Therefore, according to an exemplary embodiment,

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the distribution of one (1) in the parity-check matrix is determined by the density evolution method. That is, a degree distribution having the highest probability for error probability to converge to a predetermined value by performing iterative decoding as many as a predetermined times, that is, a degree distribution having the lowest SNR is selected from all the possible degree distributions.

In this case, restrictions are as follows.

- 1) The number of ones (1s) in an LDPC code affects the encoding and decoding complexity, and thus, the number of ones (1s) needs to be limited.
- 2) If the number of ones (1s) present in a column is various, the decoding complexity may be increased, and thus, a kind of the number of ones (1s) needs to be limited.

First, all the possible degree distributions (lists) are determined based on the parity-check matrix having parameters of the above Table 2 in consideration of the foregoing restrictions. Next, after a target SNR for each code rate is determined, when the LDPC codes encoded/decoded based on the parity-check matrix having the degree distributions present in the list at the target SNR by the density evolution are decoded, it is determined how many time of iteration is required for the error probability to converge to a predetermined value.

In this case, for all the degree distributions present in the list, when the error probability is converged to a predetermined value or less within the number of iteration times smaller than a specific value, the target SNR value is adjusted to be small and the density evolution for the degree distribution is performed again.

However, when the result value of the density evolution for all the degree distribution is not converged even within more iterations than the specific value, the SNR value is adjusted to be large and the density evolution for the degree distribution is performed again.

Among all the degree distributions present in the list, the degree distribution in which the error probability is converged to the predetermined value or less within the low SNR and the small iterative decoding (that is, iteration number of times) is determined as the result of step 1.

Step 3) The position of the non-zero quasi-cyclic matrix is determined based on the degree distribution determined in step 2.

In this step, the position of the non-zero quasi-cyclic matrix is determined based on a well known PEG algorithm (X.-Y. Hu, E. Eleftheriou, and D.-M. Arnold, "Regular and irregular progressive edge-growth tanner graphs", IEEE Trans. Inf. Theory, vol. 51, no. 1, pp. 386-398, January 2005.) In this case, additional restrictions suggested by the present invention are to make the number of 1s in each row maximally uniform.

Step 4) The indices of the quasi-cycle matrices need to be adjusted so as to prevent the error floor from occurring.

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In this step, the parity sub-matrix is fixed, and thus, is not considered. In the information word sub-matrix, only the index values of non-zero quasi-cycle matrices positioned in a predetermined row need to be changed from a column having the lowest column degree.

In this step, all index values of the non-zero quasi-cycle matrices, except the parity sub-matrix 420, positioned in the same row need to have different values.

The order of the rows at which the index values are changed may be various. A change in the index values is repeated until a cycle value of a minimum cycle is no more increased or the number of variable nodes having the minimum cycle is no more improved. Here, step 4 is called lifting.

Step 5) The parity-check matrix designed based on FIG. 4 needs to be modified into the form of FIG. 3 by the row permutation and the column permutation.

In this case, the row permutation may be performed based on following mathematical expression 9 and the column permutation may be performed based on following mathematical expression 10.

$$M \cdot i + j \Rightarrow Q_{ldpc} \cdot j + i \quad (0 \leq i < Q_{ldpc}, 0 \leq j < M) \quad (9)$$

$$i \Rightarrow i \quad (0 \leq i < K_{ldpc})$$

$$K_{ldpc} + M \cdot k + l \Rightarrow K_{ldpc} + Q_{ldpc} \cdot l + k \quad (0 \leq k < Q_{ldpc}, 0 \leq l < M) \quad (10)$$

Hereinafter, a process of designing a parity-check matrix will be described in more detail with reference to a case of a code rate 10/15 (=2/3) as an example.

Step 1) Parameters N_{ldpc} and K_{ldpc} related to the size and the values of M and Q_{ldpc} of the parity-check matrix to be designed are determined as following Table 3.

TABLE 3

code rate	N_{ldpc}	K_{ldpc}	N_{parity}	M	Q_{ldpc}	N_{qc_column}	N_{qc_row}
10/15	64800	43200	21600	360	60	180	60

Step 2) The degree distribution of the parity-check matrix is determined based on the density evolution method.

The distributions in which the error probability is highly likely to be a predetermined value at the lowest SNR as a result of the density evolution are selected from all the possible degree distributions. For example, for the selected distributions, when SNR=2.6 dB, the iteration number of times to make BER=10⁻⁵ is obtained by the density evolution method. The selected distributions and the iteration number of times obtained based on the density evolution for the selected distributions are as shown in following Table 4. In following Table 4, $N(x_i)$ represents the number of column groups or column blocks having a degree of x_i .

TABLE 4

case	Degree (x1)	Degree N(x1)	Degree (x2)	Degree N(x2)	Degree (x3)	Degree N(x3)	Degree (x4)	Degree N(x4)	The number of iterations meeting BER = 10 ⁻⁵
1	16	23	4	2	3	95	2	60	65
2	16	23	4	13	3	84	2	60	65
3	17	20	4	31	3	69	2	60	73
4	17	19	4	42	3	59	2	60	94
5	15	26	4	0	3	94	2	60	63
6	16	24	4	0	3	96	2	60	63

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Step 3) The positions of the non-zero quasi-cyclic matrix is determined based on the degree distribution determined in step 2.

According to an exemplary embodiment, a degree distribution of case 6 of above Table 4 is selected. Further, to meet the selected degree distribution, positions of the non-zero quasi-cycle matrix is determined.

The reason of determining case 6 is that, as described above, the number of ones (1s) depending on the degree distribution of the parity-check matrix affects encoding/decoding complexity, and thus, the parity-check matrix needs to be designed based on the code performance which

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is not only based on a theoretical approach called the density evolution but also on the actual encoding/decoding complexity and the designed parity-check matrix.

Step 4) The indices of the quasi-cycle matrices need to be adjusted so as to prevent the error floor from occurring.

Following Table 5 shows positions and index values of the quasi-cycle matrices of the parity-check matrix **400** having the form of FIG. 4 which is determined by the foregoing method. In this process, the method called the lifting (see foregoing step 4) considering the cycle characteristics is used.

TABLE 5

	Indices of column groups in which non-zero quasi-cycle matrix of i-th row is positioned	Indices of non-zero quasi-cycle matrix of i-th row
0	3 6 12 14 15 23 34 61 81 87 108 120 179	350 122 72 90 310 344 85 245 10 12 93 0 359
1	1 49 10 13 19 38 49 79 83 106 120 121	313 156 2 258 227 245 5 117 280 96 211 0 0
2	3 5 15 20 22 23 44 62 78 91 99 121 122	219 33 282 344 118 250 5 238 352 230 120 0 0
3	0 8 12 16 21 23 28 56 93 98 117 122 123	211 302 339 64 228 247 147 13 38 143 247 0 0
4	1 7 8 9 16 19 20 46 55 64 101 123 124	104 289 301 304 55 48 170 244 39 185 268 0 0
5	2 3 12 13 18 20 24 54 80 112 114 124 125	5 355 305 99 306 88 96 288 22 52 124 0 0
6	1 6 8 14 16 17 34 40 74 102 107 125 126	309 348 241 98 41 83 3 143 140 102 227 0 0
7	4 8 10 12 19 20 35 37 48 75 103 126 127	108 341 333 17 211 187 104 239 140 284 137 0 0
8	0 4 14 17 21 23 33 65 78 82 98 127 128	38 165 219 233 270 282 63 281 264 225 209 0 0
9	5 6 13 18 19 21 22 41 67 76 109 128 129	8 207 200 148 234 33 307 100 203 162 94 0 0
10	1 2 3 5 11 13 14 29 60 91 103 112 129 130	271 291 274 212 110 57 80 180 134 208 244 337 0 0
11	6 7 8 11 15 21 23 38 58 72 74 130 131	253 35 207 278 180 122 120 75 147 0 174 0 0
12	4 11 19 20 21 22 25 47 80 111 115 131 132	124 95 102 48 172 177 244 144 83 207 136 0 0
13	3 5 7 14 15 17 18 37 69 83 84 110 132 133	66 211 233 54 315 342 338 208 229 357 199 143 0 0
14	1 4 7 10 14 15 17 20 28 52 77 88 133 134	43 282 309 252 263 115 220 85 139 214 114 302 0 0
15	0 4 5 6 8 18 39 43 71 79 104 134 135	283 20 166 310 147 229 316 359 268 15 194 0 0
16	1 4 5 18 21 22 40 60 84 97 118 135 136	151 213 230 83 304 132 47 336 276 233 167 0 0
17	0 6 9 14 21 22 23 27 59 75 110 136 137	70 23 169 174 230 15 19 119 25 57 40 0 0
18	1 4 11 18 21 23 31 66 73 77 91 137 138	78 222 317 343 322 29 293 15 97 268 17 0 0
19	3 8 12 17 18 22 47 65 90 102 117 138 139	1 173 0 282 148 282 299 122 42 212 311 0 0
20	0 2 11 13 15 22 24 42 49 75 88 139 140	8 186 278 1 309 186 57 294 255 293 67 0 0
21	0 5 7 9 12 16 17 45 51 84 108 117 140 141	141 342 45 270 83 272 333 19 105 153 314 134 0 0
22	2 5 10 11 15 17 33 43 59 72 87 141 142	280 99 183 227 276 301 288 45 156 17 334 0 0
23	6 7 8 10 13 17 19 25 56 94 103 114 142 143	248 36 200 344 99 188 1 86 341 163 249 259 0 0
24	4 11 13 17 19 22 40 70 88 100 113 143 144	214 302 90 221 3 359 263 295 49 77 260 0 0
25	2 7 12 15 16 17 32 54 68 99 104 144 145	127 298 45 313 172 44 310 122 188 23 298 0 0
26	3 7 14 16 20 22 38 48 85 98 100 145 146	210 76 196 27 126 248 72 87 62 138 171 0 0
27	5 9 10 20 21 23 27 52 67 90 94 146 147	215 238 252 154 354 147 254 222 258 371 0 0
28	2 3 4 7 13 14 16 42 58 82 95 97 147 148	2 13 13 154 102 123 242 290 126 13 73 198 0 0
29	0 2 9 10 18 21 29 55 69 89 102 148 149	192 26 268 101 49 133 152 33 161 310 219 0 0

TABLE 5-continued

Indices of column groups in which non-zero quasi-cycle matrix of i-th row is positioned	Indices of non-zero quasi-cycle matrix of i-th row
30 1 3 8 9 10 17 20 43 63 76 100 149 150	225 53 256 42 50 265 344 15 72 65 324 0 0
31 0 2 7 9 21 22 32 47 61 92 116 150 151	251 329 4 144 203 135 90 137 242 215 71 0 0
32 3 8 12 15 16 19 23 35 64 73 106 151 152	75 9 53 305 219 76 316 278 264 41 194 0 0
33 0 1 5 6 8 9 12 36 52 72 105 112 152 153	193 289 122 345 268 45 252 142 42 244 261 216 0 0
34 2 3 6 11 13 14 30 53 83 109 119 153 154	198 238 102 316 125 311 215 304 171 233 25 0 0
35 0 5 7 17 19 23 24 63 78 96 118 154 155	325 212 281 345 46 101 151 172 236 138 51 0 0
36 2 5 9 10 12 15 19 26 56 66 71 107 155 156	55 12 237 99 228 150 213 104 91 291 92 131 0 0
37 1 4 6 18 22 23 29 51 61 85 96 156 157	210 300 236 46 341 2 333 314 107 60 342 0 0
38 2 8 11 14 16 18 19 39 54 73 82 119 157 158	110 225 136 154 222 159 50 273 159 62 240 76 0 0
39 2 9 13 15 16 20 23 36 49 86 101 158 159	125 324 179 352 157 311 41 213 260 215 224 0 0
40 1 3 6 12 18 19 31 57 92 95 114 159 160	99 265 202 210 13 265 100 118 275 280 100 0 0
41 0 4 7 9 13 20 41 50 64 74 96 160 161	266 328 333 343 93 305 316 282 57 341 293 0 0
42 1 4 14 15 19 20 28 32 86 89 110 161 162	74 103 45 205 151 222 156 106 358 170 294 0 0
43 3 10 11 16 18 21 22 45 53 58 111 162 163	148 60 91 137 330 224 338 144 128 65 284 0 0
44 2 10 11 12 13 14 15 34 57 67 68 115 163 164	21 77 206 51 84 9 316 125 322 96 210 4 0 0
45 0 1 9 10 13 20 26 59 80 106 113 164 165	310 173 346 83 239 206 327 198 262 64 280 0 0
46 2 5 7 8 15 16 19 41 44 70 85 105 165 166	250 147 233 254 87 92 243 266 13 222 227 290 0 0
47 1 4 6 12 22 23 39 53 55 94 116 166 167	110 124 212 137 199 102 123 186 133 280 231 0 0
48 0 7 10 12 16 21 22 48 66 87 97 167 168	133 207 254 348 289 340 189 35 113 273 9 0 0
49 4 5 10 11 16 23 30 51 62 90 101 168 169	290 18 27 182 317 248 4 19 294 206 311 0 0
50 3 5 14 18 19 22 25 57 89 105 119 169 170	304 298 256 329 99 110 263 181 249 340 13 0 0
51 1 8 9 13 14 16 18 42 71 81 109 170 171	102 90 110 343 80 310 19 327 250 173 93 0 0
52 0 10 11 13 16 20 35 44 86 108 115 171 172	269 356 84 303 323 200 351 49 287 307 200 0 0
53 2 11 17 21 22 23 31 63 93 104 113 172 173	74 127 272 61 163 349 261 183 125 99 117 0 0
54 0 5 6 9 11 13 45 46 65 70 81 173 174	359 30 277 163 179 218 178 286 85 286 327 0 0
55 2 6 8 17 20 23 26 37 62 76 95 174 175	280 8 87 185 136 193 300 25 261 340 120 0 0
56 3 6 8 10 19 21 36 50 60 93 116 175 176	72 89 240 39 250 214 225 22 176 88 291 0 0
57 4 6 7 9 11 12 15 33 68 69 107 176 177	73 65 150 80 133 233 273 228 4 232 6 0 0
58 0 3 7 14 15 17 18 27 46 79 92 118 177 178	124 39 149 116 100 145 240 290 57 304 105 13 0 0
59 1 12 17 18 20 21 30 50 77 99 111 178 179	317 35 148 6 341 32 347 14 326 342 237 0 0
60 3 6 12 14 15 23 34 61 81 87 108 120 179	350 122 72 90 310 344 85 245 10 12 93 0 359
61 1 4 9 10 13 19 38 49 79 83 106 120 121	313 156 2 258 227 245 5 117 280 96 211 0 0
62 3 5 15 20 22 23 44 62 78 91 99 121 122	219 33 282 344 118 250 5 238 352 230 120 0 0
63 0 8 12 16 21 23 28 56 93 98 117 122 123	211 302 339 64 228 247 147 13 38 143 247 0 0
64 1 7 8 9 16 19 20 46 55 64 101 123 124	104 289 301 304 55 48 170 244 39 185 268 0 0
65 2 3 12 13 18 20 24 54 80 112 114 124 125	5 355 305 99 306 88 96 288 22 52 124 0 0
66 1 6 8 14 16 17 34 40 74 102 107 125 126	309 348 241 98 41 83 3 143 140 102 227 0 0
67 4 8 10 12 19 20 35 37 48 75 103 126 127	108 341 333 17 211 187 104 239 140 284 137 0 0

TABLE 5-continued

Indices of column groups in which non-zero quasi-cycle matrix of i-th row is positioned	Indices of non-zero quasi-cycle matrix of i-th row
68 0 4 14 17 21 23 33 65 78 82 98 127 128	38 165 219 233 270 282 63 281 264 225 209 0 0
69 5 6 13 18 19 21 22 41 67 76 109 128 129	8 207 200 148 234 33 307 100 203 162 94 0 0
70 1 2 3 5 11 13 14 29 60 91 103 112 129 130	271 291 274 212 110 57 80 180 134 208 244 337 0 0
71 6 7 8 11 15 21 23 38 58 72 74 130 131	253 35 207 278 180 122 120 75 147 0 174 0 0
72 4 11 19 20 21 22 25 47 80 111 115 131 132	124 95 102 48 172 177 244 144 83 207 136 0 0
73 3 5 7 14 15 17 18 37 69 83 84 110 132 133	66 211 233 54 315 342 338 208 229 357 199 143 0 0
74 1 4 7 10 14 15 17 20 28 52 77 88 133 134	43 282 309 252 263 115 220 85 139 214 114 302 0 0
75 0 4 5 6 8 18 39 43 71 79 104 134 135	283 20 166 310 147 229 316 359 268 15 194 0 0
76 1 4 5 18 21 22 40 60 84 97 118 135 136	151 213 230 83 304 132 47 336 276 233 167 0 0
77 0 6 9 14 21 22 23 27 59 75 110 136 137	70 23 169 174 230 15 19 119 25 57 40 0 0
78 1 4 11 18 21 23 31 66 73 77 91 137 138	78 222 317 343 322 29 293 15 97 268 17 0 0
79 3 8 12 17 18 22 47 65 90 102 117 138 139	1 173 0 282 148 282 299 122 42 212 311 0 0
80 0 2 11 13 15 22 24 42 49 75 88 139 140	8 186 278 1 309 186 57 294 255 293 67 0 0
81 0 5 7 9 12 16 17 45 51 84 108 117 140 141	141 342 45 270 83 272 333 19 105 153 314 134 0 0
82 2 5 10 11 15 17 33 43 59 72 87 141 142	280 99 183 227 276 301 288 45 156 17 334 0 0
83 6 7 8 10 13 17 19 25 56 94 103 114 142 143	248 36 200 344 99 188 1 86 341 163 249 259 0 0
84 4 11 13 17 19 22 40 70 88 100 113 143 144	214 302 90 221 3 359 263 295 49 77 260 0 0
85 2 7 12 15 16 17 32 54 68 99 104 144 145	127 298 45 313 172 44 310 122 188 23 298 0 0
86 3 7 14 16 20 22 38 48 85 98 100 145 146	210 76 196 27 126 248 72 87 62 138 171 0 0
87 5 9 10 20 21 23 27 52 67 90 94 146 147	215 238 252 154 354 147 254 222 258 371 0 0
88 2 3 4 7 13 14 16 42 58 82 95 97 147 148	2 13 13 154 102 123 242 290 126 13 73 198 0 0
89 0 2 9 10 18 21 29 55 69 89 102 148 149	192 26 268 101 49 133 152 33 161 310 219 0 0
90 1 3 8 9 10 17 20 43 63 76 100 149 150	225 53 256 42 50 265 344 15 72 65 324 0 0
91 0 2 7 9 21 22 32 47 61 92 116 150 151	251 329 4 144 203 135 90 137 242 215 71 0 0
92 3 8 12 15 16 19 23 35 64 73 106 151 152	75 9 53 305 219 76 316 278 264 41 194 0 0
93 0 1 5 6 8 9 12 36 52 72 105 112 152 153	193 289 122 345 268 45 252 142 42 244 261 216 0 0
94 2 3 6 11 13 14 30 53 83 109 119 153 154	198 238 102 316 125 311 215 304 171 233 25 0 0
95 0 5 7 17 19 23 24 63 78 96 118 154 155	325 212 281 345 46 101 151 172 236 138 51 0 0
96 2 5 9 10 12 15 19 26 56 66 71 107 155 156	55 12 237 99 228 150 213 104 91 291 92 131 0 0
97 1 4 6 18 22 23 29 51 61 85 96 156 157	210 300 236 46 341 2 333 314 107 60 342 0 0
98 2 8 11 14 16 18 19 39 54 73 82 119 157 158	110 225 136 154 222 159 50 273 159 62 240 76 0 0
99 2 9 13 15 16 20 23 36 49 86 101 158 159	125 324 179 352 157 311 41 213 260 215 224 0 0
100 1 3 6 12 18 19 31 57 92 95 114 159 160	99 265 202 210 13 265 100 118 275 280 100 0 0
101 0 4 7 9 13 20 41 50 64 74 96 160 161	266 328 333 343 93 305 316 282 57 341 293 0 0
102 1 4 14 15 19 20 28 32 86 89 110 161 162	74 103 45 205 151 222 156 106 358 170 294 0 0
103 3 10 11 16 18 21 22 45 53 58 111 162 163	148 60 91 137 330 224 338 144 128 65 284 0 0
104 2 10 11 12 13 14 15 34 57 67 68 115 163 164	21 77 206 51 84 9 316 125 322 96 210 4 0 0
105 0 1 9 10 13 20 26 59 80 106 113 164 165	310 173 346 83 239 206 327 198 262 64 280 0 0

TABLE 5-continued

Indices of column groups in which non-zero quasi-cycle matrix of i-th row is positioned	Indices of non-zero quasi-cycle matrix of i-th row
106 2 5 7 8 15 16 19 41 44 70 85 105 165 166	250 147 233 254 87 92 243 266 13 222 227 290 0 0
107 1 4 6 12 22 23 39 53 55 94 116 166 167	110 124 212 137 199 102 123 186 133 280 231 0 0
108 0 7 10 12 16 21 22 48 66 87 97 167 168	133 207 254 348 289 340 189 35 113 273 9 0 0
109 4 5 10 11 16 23 30 51 62 90 101 168 169	290 18 27 182 317 248 4 19 294 206 311 0 0
110 3 5 14 18 19 22 25 57 89 105 119 169 170	304 298 256 329 99 110 263 181 249 340 13 0 0
111 1 8 9 13 14 16 18 42 71 81 109 170 171	102 90 110 343 80 310 19 327 250 173 93 0 0
112 0 10 11 13 16 20 35 44 86 108 115 171 172	269 356 84 303 323 200 351 49 287 307 200 0 0
113 2 11 17 21 22 23 31 63 93 104 113 172 173	74 127 272 61 163 349 261 183 125 99 117 0 0
114 0 5 6 9 11 13 45 46 65 70 81 173 174	359 30 277 163 179 218 178 286 85 286 327 0 0
115 2 6 8 17 20 23 26 37 62 76 95 174 175	280 8 87 185 136 193 300 25 261 340 120 0 0
116 3 6 8 10 19 21 36 50 60 93 116 175 176	72 89 240 39 250 214 225 22 176 88 291 0 0
117 4 6 7 9 11 12 15 33 68 69 107 176 177	73 65 150 80 133 233 273 228 4 232 6 0 0
118 0 3 7 14 15 17 18 27 46 79 92 118 177 178	124 39 149 116 100 145 240 290 57 304 105 13 0 0
119 1 12 17 18 20 21 30 50 77 99 111 178 179	317 35 148 6 341 32 347 14 326 342 237 0 0

Step 5) The parity-check matrix designed based on FIG. 4 in step 4 needs to be modified into the form of FIG. 3 by the row permutation and the column permutation.

Hereinafter, examples of the parity-check matrix designed by the foregoing method for designing a parity-check matrix will be described.

It is well known that any parity-check matrix H_1 is equivalent to a parity-check matrix H_2 obtained by performing row permutation and column permutation on the parity-check matrix H_1 . That is, any parity-check matrix having the form of FIG. 3 is equivalent to a parity-check matrix having the form of FIG. 4 obtained by performing row permutation and column permutation on the parity-check matrix of FIG. 3.

Further, a parity-check matrix having the form of FIG. 3 and a parity-check matrix, in which an order of column groups of the parity-check matrix of FIG. 3 according to the exemplary embodiment is changed, are also equivalent to each other. Further, when index values of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group are $k_1, k_2, \dots, K_m$ (herein, m is a column degree of the i-th column group), parity-check matrices are equivalent to each other even when the index values of the row at which one (1) is positioned in the zero (0)-th column of the i-th column group are changed to $(K_x + Q_{ldpc} \times y) \bmod (N_{ldpc} - K_{ldpc})$ ($1 \leq k \leq m$). Q_{ldpc} is $(N_{ldpc} - K_{ldpc}) / M$ as described above, in which M represents the number of columns belonging to a column group. Further, y is any constant.

As a specific example, in the case in which N_{ldpc} is 30, K_{ldpc} is 15, and Q_{ldpc} is 3, if a case in which position information about rows at which weight-1s are positioned in zero (0)-th columns of three column groups is represented as following Table 6,

TABLE 6

1 2 8 10
0 9 13
0 14

When y is 2,

since $1 \rightarrow (1+3 \times 2) \bmod 15 = 7$

$2 \rightarrow (2+3 \times 2) \bmod 15 = 8$

$8 \rightarrow (8+3 \times 2) \bmod 15 = 14$

$10 \rightarrow (10+3 \times 2) \bmod 15 = 1$

$0 \rightarrow (0+3 \times 2) \bmod 15 = 6$

$9 \rightarrow (9+3 \times 2) \bmod 15 = 0$

$13 \rightarrow (13+3 \times 2) \bmod 15 = 4$

$0 \rightarrow (0+3 \times 2) \bmod 15 = 6$

$14 \rightarrow (14+3 \times 2) \bmod 15 = 5,$

the parity-check matrices are equivalent to one another even when the position information about the row at which the weight-1s are positioned in the zero (0)-th columns of the three column groups is represented as following Table 7.

TABLE 7

7 8 14 1
6 0 4
6 5

Further, the parity-check matrices are equivalent to one another even when an order of the column groups is

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changed, and thus, the parity-check matrices represented as the following Table 8 are also equivalent to one another.

TABLE 8

0 9 3
0 14
1 2 3 10

Further, in the case of the parity-check matrix having the form of FIG. 4 according to the exemplary embodiment, when the index values of the quasi-cycle matrix present in the same column group or column block are added or

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subtracted to or from the same value, the parity-check matrix may be equal to the previous parity-check matrix. Further, the parity-check matrices in which an order of the column-blocks is changed are equivalent to one another.

Hereinafter, examples of a parity-check matrix having the form of FIG. 3 designed by the method for designing a parity-check matrix described in the exemplary embodiment will be described.

As an example, if a codeword length N_{ldpc} is 64800, a code rate R is $6/15$, M is 360 and Q_{ldpc} is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 9.

TABLE 9

i	Indices of rows at which 1s are positioned in 0-th column of i -th column group
0	891 2309 3615 5472 8525 10308 11096 11503 11553 12420 13298 15489 15883 17561 21293 23132 25133 28163 30811 31810 34215 34584 36541 38410
1	3290 5479 5596 6546 7021 7566 9274 9303 10047 10278 10956 12393 18309 19755 19790 20828 22690 25235 26853 31571 31893 36500 36772 37536
2	1031 4765 5011 5200 6189 8985 12056 12366 14439 15482 16521 18338 19165 19476 21455 22096 22589 23699 23838 24649 30113 31490 31960 34564
3	1619 2364 2889 3409 7278 7384 8220 11204 11694 12064 16087 16722 19633 21525 24373 25817 29614 31392 33508 34329 34352 37006 37080 38444
4	3612 5474 6202 8521 11153 12202 12213 12962 13868 14359 14385 15072 15375 18156 18308 26375 28004 28905 29697 30000 30686 32819 34407 38688
5	1441 2527 3103 3297 6303 6421 7233 8011 8250 8909 10558 13855 14699 17508 17810 21259 22423 23264 29249 30830 32607 35202 38303 38402
6	4 515 1864 6876 7245 7371 9190 11814 13748 15774 20161 21526 24462 24884 25737 26251 27266 29957 34779 35082 36664 37354 37516 37837
7	368 4867 6985 7663 15368 15947 16235 16544 17468 18936 21544 21903 22422 23080 25411 25938 26128 29554 29847 31845 32714 36101 37324 38004
8	1125 2843 3161 3172 6296 8216 9506 9822 10005 11459 12555 12919 13814 13872 17916 19481 22046 23394 26020 32002 35467 36616 38306 38784
9	803 1133 1439 4618 5708 7048 8114 8195 9089 10238 12403 12520 14005 14194 16527 19186 20343 22595 27314 30831 35338 36223 36754 38504
10	988 1723 2736 2849 3827 7076 12613 12989 13268 14323 15013 20703 20872 21170 25384 29187 29524 29693 30029 30948 32598 32733 32906 34586
11	6085 8070 8611 10860 11848 13126 15737 19744 21503 21572 21598 22468 23444 23753 26696 26722 29679 29999 31803 32206 33479 34221 36608 38731
12	1410 1790 3290 7470 9833 10001 10966 11262 11687 12155 12807 16606 17295 20105 23354 23405 27107 28277 29386 30326 32151 32641 33649 36765
13	817 2781 2937 3886 5881 9657 10675 10816 11934 13044 16160 19321 19567 24463 26365 28182 29386 31428 31708 33113 33975 34629 36265 36650
14	1393 6471 7153 7266 7647 11124 13430 16396 21033 21227 23273 24502 25273 27888 29107 30726 31551 31987 32737 33126 33764 35951 36021 36264
15	5003 10529 15986 32408
16	1236 7663 14599 24745
17	10824 16168 16783 29318
18	6834 29474 29608 29779
19	2709 8594 20859 23739
20	6014 11516 14715 36702
21	2784 6493 27798 35084
22	1044 5892 13056 18720
23	2875 19737 21893 32447
24	11358 17602 34706 38472
25	11527 23325 28196 33428
26	2428 2597 3177 3909
27	29269 33353 33791 37487
28	204 14925 22290 26379
29	18959 19007 24212 27094
30	13813 14631 17097 31294
31	11533 17559 30617 37966
32	14821 22942 28281 28353
33	3440 8417 8534 24133
34	10186 20333 24737 34733
35	12388 31132 33131 36927
36	7311 16914 19363 30928
37	16762 23990 30952 36068
38	1872 9840 33270 37616
39	14047 17094 18244 24107
40	1903 21121 22616 38860
41	5979 16278 27327 34338
42	7644 10438 25622 36926
43	1243 27018 31393 38210
44	179 1944 5719 15319
45	17318 24311 37325

TABLE 9-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group		
46	1528	10135	33640
47	1321	15978	25776
48	16957	21572	26719
49	14349	15799	29164
50	574	5582	20145
51	8814	14785	31081
52	667	7530	25659
53	7823	23615	37483
54	12766	30755	35696
55	4302	11660	36217
56	7393	11724	20441
57	864	13910	22924
58	5500	10085	31057
59	5399	13946	32583
60	1364	12423	36203
61	12356	24958	32026
62	10894	33813	38525
63	15456	15794	28350
64	6603	25570	33797
65	8709	19457	38831
66	330	9502	24362
67	12579	15733	34408
68	28443	34954	37293
69	3123	8513	19702
70	16976	22876	26384
71	20399	29971	33952

That is, positions of a row at which one (1) is positioned in a zero (0)-th column of each column group may be defined by indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group shown in above Table 9, and, by shifting a row at which one (1) is positioned in a zero (0)-th column of each column group by Q_{ldpc} , rows at which one (1) is positioned in other columns of the column group may be defined.

In the foregoing example, since $Q_{ldpc}=(64800-25920)/360=108$ and the indices of the row at which one (1) is positioned in the zero (0)-th column of the zero (0)-th column group are 891, 2309, 3615, . . . , the indices of the row at which one (1) is positioned in the 1-th column of the

zero (0)-th column group may be 999 ($=891+108$), 2417 ($=2309+108$), 3723 ($=3615+108$), . . . and the indices of the row at which one (1) is positioned in the second column of the zero (0)-th column group may be 1107 ($=999+108$), 2525 ($=2417+108$), 3831 ($=3723+108$),

A parity-check matrix may also be defined by the method described in above Table 9, according to an exemplary embodiment.

As another example, if a codeword length N_{ldpc} is 64800, the code rate R is 8/15, M is 360 and Q_{ldpc} is 84, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 10.

TABLE 10

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group														
0	575	3480	5049	7421	9300	10191	12344	12662	13287	13828	15934	17773	17905	18805	20237
1	4644	4943	6642	8110	9277	12828	13067	13101	13659	17652	17759	18222	18901	19452	21242
2	4600	5990	11592	11782	12150	13399	16582	17315	18200	18446	21324	21563	22895	23558	24829
3	727	4247	4937	8089	15745	18832	19983	20501	20832	21763	21809	22581	24885	25564	26204
4	2596	2687	3647	5652	7705	7911	7994	9284	10109	13037	14671	15837	17416	18970	19791
5	1056	3435	6648	7544	8563	10425	13131	13942	17003	19076	20132	23804	25730	25979	26464
6	1410	2632	3806	7856	8285	8683	8968	10120	12354	16212	16354	18750	20643	23389	23593
7	785	1301	2206	3442	6642	8532	10226	10372	12226	15724	17648	19961	20314	21981	25084
8	957	983	1450	3635	4882	5169	5906	8105	10539	11404	13286	14847	15216	15238	15705
9	465	2753	4029	5584	6626	7620	7688	8867	9205	9786	10661	11396	12721	12757	15181
10	2018	3017	5775	7512	8926	10258	11871	13763	14732	15346	17677	17762	18625	21929	22892
11	1686	2538	3165	3716	10989	11643	13764	14063	15621	15940	16710	16739	17442	18607	18661
12	983	2443	3139	8694	8930	11798	12540	13629	14446	16902	18391	18452	19745	23125	24267
13	744	4121	4252	9986	10475	13103	13185	14142	15345	15352	15445	17573	21329	21790	21879

TABLE 10-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
14	1468 4636 4903 6223 7505 7748 7986 9273 11609 11821 13188 18694 19231 20890 21722 23933 25493 26809 29300
15	295 5207 5830 5967 9076 9476 10287 14563 14863 16497 16877 17122 17132 21298 23456 23941 24040 25165 29292
16	6219 6354 6512 6678 7214 7968 9600 15508 15635 17351 17825 18054 18545 21172 21604 25263 28503 29776 29957
17	1399 1866 4118 6022 6498 8718 8929 10364 10908 11588 13031 17426 19049 20238 23750 25217 25437 27795 30143
18	1263 1605 3107 5328 5522 7461 7484 9206 10185 11691 12614 12955 15109 22915 24383 26051 27802 27882 29485
19	684 1033 2283 3002 4245 4663 8887 9768 12403 14232 16792 18157 18826 21931 22730 25789 25975 26985 29162
20	2377 18147 29547 29698
21	3083 4540 6201 17778
22	468 9283 14739 24011
23	4848 12149 14672 26446
24	8851 9316 12511 18952
25	14327 21539 22753 28982
26	169 5297 12557 15108
27	8941 20588 23889 29192
28	17157 17418 24573 26133
29	7756 8650 28174 28806
30	8550 12638 19454 27484
31	15398 23410 25520 26808
32	7634 15554 19352 27115
33	6090 13925 17827 29331
34	1845 2352 5982 19601
35	471 1464 3191 3634
36	17099 18086 23525
37	3397 15058 30224
38	4864 25880 26268
39	1096 4775 7912
40	1314 3259 17301
41	2481 8396 15132
42	17825 28119 28676
43	2343 8382 28840
44	18374 20939 27091
45	1290 8786 15916
46	1481 4710 28846
47	2185 3705 27086
48	5496 15681 21854
49	12697 13407 22178
50	12788 21227 22894
51	2854 6232 8609
52	2289 18227 27458
53	1965 21935 23001
54	3836 7081 12282
55	1976 18845 23135
56	497 9717 26670
57	22046 27327 30067
58	12068 28045 28990
59	2023 10933 16444
60	19566 23905 25186
61	13303 13834 28813
62	10572 20305 21388
63	14093 18024 24286
64	3612 21383 23582
65	50 11267 12288
66	771 5652 27795
67	16131 20047 25649
68	13227 23035 24450
69	4839 13467 27488
70	2852 4677 21481
71	2504 4680 15664
72	12518 14518 24267
73	1222 2218 11859
74	9660 15774 18261
75	232 6424 29978
76	9750 11165 16295
77	2706 4894 28489
78	3301 14110 28612
79	2128 14436 15883
80	6274 17243 21989
81	13202 18006 22517
82	11159 16111 21608
83	3719 11563 22100
84	1756 2020 18861
85	20913 29473 30103

TABLE 10-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
86	15091 26976 27173
87	8217 9114 12963
88	5395 18516 28235
89	3859 17909 23051
90	5733 16513 18373
91	1935 3492 8437
92	11903 16760 29914
93	6091 10469 29997
94	2895 22370 29958
95	1827 12296 20070

As another example, if a codeword length N_{ldpc} is 64800, the code rate R is 10/15, M is 360 and Q_{ldpc} is 60, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 11.

TABLE 11

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	114 2135 3045 4635 5512 5681 6571 8943 10053 10109 13161 13668 14218 17417 19328 21140
1	2639 2821 3066 4293 5350 8130 9037 11265 12556 15047 15364 15531 15700 16938 17202 19034
2	1891 4150 4822 4855 6646 9754 10460 14005 14139 15038 17213 18336 20069 20384 21305 21508
3	305 600 3410 5170 5740 7354 8462 9026 12763 17132 17336 17653 18450 19318 20848 21559
4	1961 3637 4249 4694 8298 8784 8836 11708 12241 14172 14207 15127 15462 17277 20415 20848
5	1101 3770 7816 8727 8890 8915 8953 11655 12826 14313 15682 19622 19854 20569 20916 21129
6	726 933 3015 5034 6431 6743 7477 8927 9189 9520 14280 15514 16316 17757 20237 21175
7	1661 3074 3745 4264 4775 7633 7666 9228 12388 12657 12718 17066 18921 19463 19511 21391
8	1147 3483 3544 5553 6270 6406 7146 7256 8138 9191 9623 11239 12795 16251 16435 21092
9	885 1061 2199 3364 5421 5549 7347 7416 11477 11874 12991 15051 16857 18933 19110 21481
10	292 983 1627 6121 6408 6494 6507 10642 15569 15696 16665 17024 18043 18630 19316 20029
11	2598 2674 3504 4931 4940 8002 9284 10729 10914 13478 13677 14033 15010 15912 16183 16612
12	19 768 1263 3305 6513 7677 7956 9040 13427 16641 17280 18452 18584 18925 19559 20587
13	1071 3472 7305 7981 8574 9609 10899 14134 15508 15665 15683 16061 16224 16604 18190 21560
14	2974 5834 6290 8468 9866 11177 12398 14248 14698 15726 16200 16810 16851 18373 18942 21104
15	519 2684 2713 2845 3000 3080 3332 4682 5062 5277 9342 10811 12636 14714 15658 16426
16	2272 2629 3051 4308 5301 7108 8318 8492 11305 12219 13423 16126 17763 18304 19146 20006
17	935 1093 1641 3562 4699 5333 5730 7628 8364 8414 10343 10555 12779 12958 16626 18985
18	1038 1333 1843 1910 3245 7258 7875 12098 12729 12739 16636 18689 18877 20511 20860 21299
19	5740 6656 6901 7066 7569 8856 8947 12582 15492 15710 17072 18638 18724 18875 21444 21563
20	962 990 1199 2979 3341 8322 9285 9652 10387 11404 12387 13495 14066 16325 16514 18732
21	387 1248 2298 3376 5408 7817 7923 8203 8816 9451 11292 13649 14291 17993 19629 19739
22	84 1177 1363 3189 4699 6746 9707 10308 10460 10992 11873 13531 13696 14522 15050 20717
23	713 960 2672 4688 6602 6769 6783 10075 12807 14411 15527 15575 19179 19878 20477 21517
24	12575 15845 18200
25	5870 6972 16463
26	2025 3655 15396
27	4258 6387 14477
28	12282 12783 13274
29	1657 10810 12509
30	839 8734 21409
31	4038 5993 15640

TABLE 11-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
32	3025 15282 16231
33	4342 7977 17828
34	14144 16500 21426
35	592 4952 15367
36	8156 8859 13113
37	7267 9133 20155
38	17111 17306 21301
39	2655 5258 14267
40	5844 13026 18796
41	2681 5686 15609
42	2031 3980 4228
43	75 18922 20730
44	18712 20866 21302
45	10974 13003 20481
46	4494 6964 18238
47	3679 12972 13411
48	13207 16406 19548
49	6039 6320 14581
50	4721 20336 20819
51	2797 15321 20509
52	8307 8774 19113
53	3394 10487 13963
54	4325 12098 14305
55	13667 19264 19649
56	1163 16176 20823
57	2324 10790 14560
58	12791 14068 17743
59	9765 12262 20117
60	1456 11096 13570
61	6900 7111 15217
62	4009 5995 7322
63	10673 11315 17310
64	5792 10504 18221
65	4748 14299 16554
66	4176 14868 20718
67	6147 9429 15884
68	9044 10345 21417
69	7737 7873 11969
70	3924 4494 8326
71	5535 6651 16116
72	11 6993 20602
73	15798 17918 19172
74	1181 11171 13206
75	4040 4567 18197
76	1255 11889 17730
77	2099 5538 14774
78	482 5768 7475
79	3418 4801 20715
80	5925 16632 20285
81	2034 11271 21000
82	7238 8108 20848
83	193 11374 15841
84	5056 9673 12441
85	8026 17906 18037
86	162 4432 8739
87	1582 5268 20880
88	3494 17600 18684
89	3029 6710 11442
90	9289 19099 19407
91	7802 9130 20598
92	5140 8731 15358
93	14153 16376 19323
94	4847 11843 21567
95	4840 14455 17248
96	1117 4061 13355
97	7636 9748 21108
98	9068 13023 13346
99	1139 14402 20245
100	2190 11366 17004
101	2989 5524 8199
102	8489 8899 15486
103	6683 6970 13387
104	3745 9975 15713
105	1250 4246 5973
106	8941 9992 17805
107	7986 13776 21297
108	2781 3232 16020
109	7654 15969 16071

TABLE 11-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
110	4002 13033 19217
111	4603 7439 9192
112	1390 8673 18485
113	4845 6024 14633
114	6083 14165 15640
115	9652 13452 21404
116	4196 7787 17371
117	2959 6783 13581
118	11596 18575 20878
119	17078 20134 20870

As another example, if a codeword length N_{ldpc} is 64800, the code rate R is 12/15, M is 360 and Q_{ldpc} is 36, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 12.

TABLE 12

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	57 3127 3324 3362 3471 4108 4850 5588 5721 6436 6845 9325 11745 12956
1	651 893 1879 2641 2807 4123 4406 5364 6346 8530 9387 10301 11047 12660
2	1925 2592 2655 4108 4433 4544 4614 7081 7254 7759 7815 9121 9453 10613
3	718 1559 1859 2697 3739 3948 4506 4982 6898 7410 7569 8243 10840 12075
4	21 1387 1645 1980 4188 4751 6148 7864 8498 8647 9715 10513 11114 12910
5	1082 2045 2395 3663 4597 4727 7055 8123 8189 8618 8782 11114 11716 12211
6	312 951 2269 2569 4820 4972 5803 5816 6406 6861 8277 8398 9349 10816
7	20 1833 2300 4962 5267 5653 6215 7098 7434 7509 9235 10331 10731 11060
8	594 1091 1743 1809 2686 3095 3316 5381 5927 6115 7100 7750 9901 11091
9	1065 1236 1716 2628 4589 5164 5209 6502 7719 10107 11333 11599 12011 12893
10	1799 2958 3999 4196 4351 4358 5019 5128 5840 6154 6961 9328 10530 11440
11	2982 3133 4612 4776 4986 6276 6992 7372 7589 7831 8109 8619 11594 12694
12	282 3140 3300 4219 4459 5189 5525 6753 8154 8522 8931 9073 10019 10654
13	298 1137 3973 4000 7328 7923 8602 8734 8792 9419 9748 9988 10200 10985
14	1142 1639 1997 3951 4388 4638 5676 6478 6738 6927 8643 8793 11528 11949
15	629 1735 3072 3268 3585 3888 4491 4622 5294 6817 6829 9003 9930 10426
16	486 1200 1594 2713 3273 3440 4667 5247 7304 7901 8302 8576 9014 10180
17	2377 2471 3742 5186 7137 9518 9798 10029 10060 11052 12109 12660 12716 12762
18	175 1242 2118 4101 4130 6252 6598 7272 7342 7834 7859 7924 10940 11032
19	550 1349 2360 2373 2895 3442 5258 5695 8512 8882 10710 11562 11940 12432
20	1791 2788 4958 5573 5917 5952 6195 7376 7920 7971 8983 10817 10883 11550
21	639 1000 2387 2429 3548 3659 4186 6741 7490 8383 8760 9392 9649 11971
22	523 1269 2262 2987 3073 3811 4227 5028 5181 8471 8712 9421 9509 11871
23	117 266 505 1967 2620 3188 5282 7194 8119 8251 8290 10433 12174 12337
24	896 3196 3714 4979 5258 5584 5945 7142 7713 9745 9789 11252 11339 11945
25	864 2383 2594 3398 5969 6116 6418 6649 7199 7242 8169 9221 9740 9890
26	1334 1487 12881
27	1447 4633 11032
28	273 1257 10453
29	5102 10999 12496
30	7578 9743 10513
31	7156 10266 10546
32	3199 7369 7824
33	2354 7674 8569
34	3687 9478 12663
35	2997 9418 9581
36	6519 8081 11229
37	9894 9899 11254
38	6 9344 11731
39	9612 12680 12776
40	353 10896 11643
41	945 2537 9804
42	2915 8984 11098
43	2353 4884 7456
44	6003 8924 11646
45	349 748 8625
46	4799 7204 12240
47	2464 8958 11020
48	1915 2903 12358
49	2246 3032 12531
50	2594 12742 12914
51	2002 10995 12079
52	853 1049 5022
53	4142 4301 6413
54	914 3882 12047

TABLE 12-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
55	5479 10413 11225
56	228 6874 11183
57	2836 9737 10728
58	1795 9981 12734
59	2641 2844 11779
60	1245 9983 12804
61	6002 8612 9704
62	1237 1760 7504
63	844 1485 5869
64	2657 4461 5642
65	2423 4203 9111
66	244 1855 6131
67	5318 6371 11430
68	391 1617 10126
69	1762 9259 10603
70	2604 4335 6702
71	4381 5486 8045
72	7667 8875 11451
73	1968 4023 6911
74	4630 10184 11357
75	6582 12348 12769
76	3840 6302 7388
77	1 4197 5358
78	1265 3153 11352
79	2504 7180 10044
80	1980 5027 9717
81	5699 6899 9668
82	1432 2803 3314
83	1237 8470 9642
84	829 6745 7923
85	329 1931 5575
86	1067 6867 7257
87	4744 5559 7835
88	109 6756 12238
89	2814 5237 11153
90	1592 10696 10749
91	225 2293 6260
92	646 2170 7578
93	6466 8222 9838
94	44 6574 12160
95	1755 2734 12780
96	1249 8264 8318
97	5789 6622 9481
98	5666 8681 8744
99	123 5803 9291
100	3750 7919 9167
101	1064 2848 12753
102	2142 8656 9244
103	6193 7219 11732
104	7356 7819 9928
105	4780 5937 11993
106	5092 7186 9141
107	1238 3840 12360
108	1649 2096 2587
109	5560 5903 12899
110	1134 4341 8330
111	1645 9495 10041
112	9585 11595 12912
113	8748 10646 11894
114	45 7255 9074
115	1051 2694 6188
116	622 3460 8394
117	3598 4623 9025
118	1218 3540 12843
119	4938 8698 12423
120	1766 3635 11427
121	5177 6706 9127
122	943 3590 10245
123	4864 7394 11117
124	5852 6042 10421
125	8285 10775 12349
126	787 7171 7866
127	718 4688 12234
128	728 2353 10667
129	3629 4592 6485
130	2880 5157 11466
131	2906 10220 11796
132	4243 5440 10907

TABLE 12-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group			
133	5262	7543	12303	
134	4440	7779	10940	
135	2515	5843	9202	
136	4684	8874	10586	
137	2270	7197	8652	
138	7190	7870	8317	
139	1158	10456	12909	
140	1583	7669	10781	
141	8141	11209	12555	
142	3181	3903	7832	
143	2428	4467	8074	

Meanwhile, in the design process of codes, the process of step 4 uses the lifting method considering the cycle characteristics. When various algebraic characteristics as well as the cycle characteristics are additionally considered, codes having better performance may also be designed.

Generally, since the performance of the LDPC codes affects the degree distribution as well as the cycle characteristics, codes having better performance may be designed in consideration of both of the two characteristics.

According to an exemplary embodiment, parameters may be determined depending on cycles having the shortest length and the number of variable nodes included in these cycles and having a specific degree, and a sequence may be determined depending on a rule determined based on the parameters in a lifting step.

For example, when the lifting process is applied to a column group of which the degrees are A and B, only a cycle in which the number of variable nodes of which the degree is A among the variable nodes included in the cycle while the lifting of the column group of which the degree is A is performed is x_1 or less and a value of the variable nodes (the number of variable nodes of which the degree is A+the number of variable nodes of which the degree is BxC) included in the cycle while the lifting of the column group of which the degree is B is x_2 or less may be considered.

However, all cycles are considered when the lifting process is applied to the column groups of which the degrees are not A and B.

Here, A is a positive integer of two (2) or more, B is a positive integer larger than A and smaller than a maximum degree of the parity-check matrix, and C is a weighting factor and has a positive integer value.

In addition, all cycles are considered without distinction of a specific cycle when the lifting process is applied to the column groups of which the degrees are not A and B.

As a specific example, the case in which A=3, B=4, and C=2 will be described.

For example, in the lifting process, the lifting is performed in consideration of only cycles in which the number of variable nodes (which is equivalent to columns) of which the degree is three (3) among the variable nodes included in

the cycles in the process of performing the lifting of column groups of which the degrees are three (3) is x_1 or less and (the number of variable nodes of which the degree is $4 \times 2 +$ the number of variable nodes of which the degree is three (3)) among variable nodes included in the cycles in the process of performing the lifting of column groups of which the degree is four (4) is x_2 or less.

That is, when the index values are changed in the lifting process, the lifting process is repeated until the number of variable nodes having the cycles is not improved to change the index values depending on the degrees of each variable node in consideration of only the cycles corresponding to the conditions.

Generally, in simple lifting, a parity-check matrix is designed so that the number of cycles is decreased. In this process, the parity-check matrix is designed such that the number of cycles meeting the above-mentioned conditions is decreased.

The reason why the above-mentioned conditions affect the design of LDPC codes having better performance is that the cycles and the degree distributions affecting the LDPC codes have been considered in the above-mentioned conditions.

Examples of the LDPC codes designed in consideration of these additional conditions are shown in following Tables 13 to 18. For reference, the same length and code rate as those of the LDPC codes as shown in above Tables 9 to 12 are designed, values (x_1, x_2) considered in following Table 13 are equal to (4, 5), values (x_1, x_2) considered in following Table 14 is equal to (4, 5), (x_1, x_2) considered in following Table 15 are equal to (3, 4), and values (x_1, x_2) considered in following Table 16 are equal to (3, 4). Further, values (x_1, x_2) considered in following Table 17 are equal to (8, 8) and (x_1, x_2) considered in following Table 18 is equal to (3, 4).

As an example of results designed in consideration of these additional conditions, if a codeword length N_{ldpc} is 64800, a code rate R is 6/15, M is 360, and Q_{ldpc} is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 13.

TABLE 13

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group															
0	253	1553	2024	4493	5350	5664	6351	8563	8803	9576	9936	11995	12166	13193	13412	17088
	21893	22167	22478	22673	23735	35313	36981	37527								
1	1238	1701	2316	3621	7222	8387	9190	10178	10599	10744	11743	18179	22465	23562		
	23820	26493	28168	28515	28940	29982	31196	32946	33687	34305						
2	1475	1874	2994	3168	3381	3988	7646	9309	11199	12856	15025	16586	20113	23155		
	24030	27491	28235	29392	30885	32896	33656	33785	34685	38713						
3	512	721	1813	3144	3276	6198	13540	14553	15017	15548	16120	17449	19101	19763		
	24180	25629	27612	29682	30910	32038	35346	36607	36836	37876						

TABLE 13-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
4	333 1124 3269 4236 4920 5207 6154 7041 7282 7979 8472 9302 10033 14295 15888 16485 18963 24572 26642 27516 30242 31209 32000 33259
5	2528 4425 4656 6631 8875 10621 12590 13334 14011 16406 16937 18942 20315 24078 24889 27298 29555 30123 30513 33101 33403 33787 36651 37975
6	414 721 898 2093 3813 8358 9316 11235 12032 12568 14339 14908 15390 19210 19450 22689 22840 28044 31218 35042 35348 35863 37611 37837
7	979 3559 4988 6900 7254 7491 11518 12297 12928 14894 15473 16179 16667 17744 20983 22854 24913 25640 28792 29536 30428 32284 33732 34523
8	353 1877 2171 5080 7140 7878 8762 11558 14836 15000 15513 16490 17423 18943 21094 22348 23394 24182 26203 28328 32408 34284 38061 38259
9	142 1961 5219 5816 9555 10358 10675 15251 15716 16079 20566 21470 23007 25419 27130 28000 28693 30742 30862 33209 33472 33851 35546 36115
10	1388 4346 5764 6052 7940 9207 13401 13603 14233 14411 16310 16598 22524 22824 26535 27965 28433 30605 31984 34241 35730 36139 37261 37720
11	106 1441 3892 4300 5026 7207 8648 14012 15828 17007 19409 22942 26363 28363 28806 29351 29722 33033 35204 35315 35824 37901 38036 38643
12	2607 5309 6506 9122 9318 9889 10322 12074 13373 14058 15341 15774 18154 18749 21949 27887 29885 31294 31487 32769 32890 37983 38403 38879
13	2154 4374 10366 10605 11179 15994 18855 20342 23936 24777 25768 26371 26745 27049 27324 27493 27985 29781 30148 33240 33673 34441 35057 35731
14	3054 6385 6561 6755 7795 9366 10392 12042 12832 14851 16187 17441 18536 20967 21792 24084 24505 26677 28167 28334 35199 36745 37533 37786
15	2735 3833 10268 17606
16	559 9607 17652 34573
17	9528 12139 14306 38416
18	11978 17094 25891 30040
19	14811 15531 27333 31274
20	12926 28602 32103 35600
21	8616 14917 18992 23478
22	7416 8568 10248 36672
23	3611 15521 17793 32467
24	4662 7908 8894 32722
25	8156 28077 32840 35719
26	977 17949 23380 26181
27	7655 9515 12185 17821
28	1107 8520 16782 23241
29	16942 19784 22031 22955
30	1054 23163 23793 37789
31	8509 17981 22306 33327
32	2253 11397 16225 30826
33	23654 30073 31421 33140
34	18286 19445 29945 38405
35	7535 8608 21915 38260
36	11082 13972 17895 18931
37	8114 25376 25660 33718
38	3380 5946 20124 37272
39	10159 18822 22456 29291
40	15289 18016 18512 37003
41	18471 19326 27834 28119
42	15326 28272 36142 37826
43	12277 20646 22411 22658
44	12199 23328 35495 36055
45	17318 24311 29009
46	1528 32815 33640
47	15978 19141 25776
48	8683 11960 16957
49	15799 20829 29164
50	574 5582 20145
51	8814 18673 31081
52	8394 25659 36739
53	23615 32879 37483
54	12766 30755 35696
55	4302 18788 25093
56	13205 14424 33529
57	864 16070 22924
58	7120 10085 31057
59	5399 13946 32583
60	1364 12423 19355
61	12356 24958 32026
62	11997 25582 36581
63	6074 9516 18414
64	7102 22587 33797
65	8709 19457 38831
66	9502 10266 24362
67	1801 12579 34408
68	471 34954 37293
69	1666 3123 8513

TABLE 13-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
70	9160 11576 26384
71	14311 20399 33952

As another example of results designed in consideration of these additional conditions, if codeword length N_{ldpc} is 64800, a code rate R is 8/15, M is 360, and Q_{ldpc} is 84,¹⁰ indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 14.

TABLE 14

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	2768 3039 4059 5856 6245 7013 8157 9341 9802 10470 11521 12083 16610 18361 20321 24601 27420 28206 29788
1	2739 8244 8891 9157 12624 12973 15534 16622 16919 18402 18780 19854 20220 20543 22306 25540 27478 27678 28053
2	1727 2268 6246 7815 9010 9556 10134 10472 11389 14599 15719 16204 17342 17666 18850 22058 25579 25860 29207
3	28 1346 3721 5565 7019 9240 12355 13109 14800 16040 16839 17369 17631 19357 19473 19891 20381 23911 29683
4	869 2450 4386 5316 6160 7107 10362 11132 11271 13149 16397 16532 17113 19894 22043 22784 27383 28615 28804
5	508 4292 5831 8559 10044 10412 11283 14810 15888 17243 17538 19903 20528 22090 22652 27235 27384 28208 28485
6	389 2248 5840 6043 7000 9054 11075 11760 12217 12565 13587 15403 19422 19528 21493 25142 27777 28566 28702
7	1015 2002 5764 6777 9346 9629 11039 11153 12690 13068 13990 16841 17702 20021 24106 26300 29332 30081 30196
8	1480 3084 3467 4401 4798 5187 7851 11368 12323 14325 14546 16360 17158 18010 21333 25612 26556 26906 27005
9	6925 8876 12392 14529 15253 15437 19226 19950 20321 23021 23651 24393 24653 26668 27205 28269 28529 29041 29292
10	2547 3404 3538 4666 5126 5468 7695 8799 14732 15072 15881 17410 18971 19609 19717 22150 24941 27908 29018
11	888 1581 2311 5511 7218 9107 10454 12252 13662 15714 15894 17025 18671 24304 25316 25556 28489 28977 29212
12	1047 1494 1718 4645 5030 6811 7868 8146 10611 15767 17682 18391 22614 23021 23763 25478 26491 29088 29757
13	59 1781 1900 3814 4121 8044 8906 9175 11156 14841 15789 16033 16755 17292 18550 19310 22505 29567 29850
14	1952 3057 4399 9476 10171 10769 11335 11569 15002 19501 20621 22642 23452 24360 25109 25290 25828 28505 29122
15	2895 3070 3437 4764 4905 6670 9244 11845 13352 13573 13975 14600 15871 17996 19672 20079 20579 25327 27958
16	612 1528 2004 4244 4599 4926 5843 7684 10122 10443 12267 14368 18413 19058 22985 24257 26202 26596 27899
17	1361 2195 4146 6708 7158 7538 9138 9998 14862 15359 16076 18925 21401 21573 22503 24146 24247 27778 29312
18	5229 6235 7134 7655 9139 13527 15408 16058 16705 18320 19909 20901 22238 22437 23654 25131 27550 28247 29903
19	697 2035 4887 5275 6909 9166 11805 15338 16381 18403 20425 20688 21547 24590 25171 26726 28848 29224 29412
20	5379 17329 22659 23062
21	11814 14759 22329 22936
22	2423 2811 10296 12727
23	8460 15260 16769 17290
24	14191 14608 29536 30187
25	7103 10069 20111 22850
26	4285 15413 26448 29069
27	548 2137 9189 10928
28	4581 7077 23382 23949
29	3942 17248 19486 27922
30	8668 10230 16922 26678
31	6158 9980 13788 28198
32	12422 16076 24206 29887
33	8778 10649 18747 22111
34	21029 22677 27150 28980
35	7918 15423 27672 27803
36	5927 18086 23525
37	3397 15058 30224
38	24016 25880 26268

TABLE 14-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
39	1096 4775 7912
40	3259 17301 20802
41	129 8396 15132
42	17825 28119 28676
43	2343 8382 28840
44	3907 18374 20939
45	1132 1290 8786
46	1481 4710 28846
47	2185 3705 26834
48	5496 15681 21854
49	12697 13407 22178
50	12788 21227 22894
51	629 2854 6232
52	2289 18227 27458
53	7593 21935 23001
54	3836 7081 12282
55	7925 18440 23135
56	497 6342 9717
57	11199 22046 30067
58	12572 28045 28990
59	1240 2023 10933
60	19566 20629 25186
61	6442 13303 28813
62	4765 10572 16180
63	552 19301 24286
64	6782 18480 21383
65	11267 12288 15758
66	771 5652 15531
67	16131 20047 25649
68	13227 23035 24450
69	4839 13467 27488
70	2852 4677 22993
71	2504 28116 29524
72	12518 17374 24267
73	1222 11859 27922
74	9660 17286 18261
75	232 11296 29978
76	9750 11165 16295
77	4894 9505 23622
78	10861 11980 14110
79	2128 15883 22836
80	6274 17243 21989
81	10866 13202 22517
82	11159 16111 21608
83	3719 18787 22100
84	1756 2020 23901
85	20913 29473 30103
86	2729 15091 26976
87	4410 8217 12963
88	5395 24564 28235
89	3859 17909 23051
90	5733 26005 29797
91	1935 3492 29773
92	11903 21380 29914
93	6091 10469 29997
94	2895 8930 15594
95	1827 10028 20070

As another example of results designed in consideration of these additional conditions, if a codeword length N_{ldpc} is 64800, a code rate R is 10/15, M is 360, and Q_{ldpc} is indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table

TABLE 15

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	501 1533 5943 9232 10258 10428 10965 11934 14081 14708 15509 17251 18380 19815 20075 20237
1	753 1266 2017 3107 4210 4770 6520 10861 11594 13191 14116 18342 18604 18825 19398 21479

TABLE 15-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
2	2379 3029 3140 3398 4528 6562 8575 9593 11196 11585 12931 13064 13825 15886 20854 21010
3	733 1262 2250 4910 8165 8374 8698 10543 10930 12940 14520 14936 15752 16879 19226 20188
4	2056 2341 4237 4807 6469 7708 8895 9548 13274 13404 13481 14082 15647 17712 19377 19638
5	2882 3081 3633 4047 4755 5094 5589 6709 12526 12710 12910 13342 14196 17836 20353 21095
6	34 309 1187 3000 3097 3246 6280 6873 7074 8935 12615 13517 14363 16317 19856 20591
7	2181 2381 3551 6904 6995 8248 9023 9348 9433 11097 12914 17326 17671 18658 19585 20846
8	915 2036 2104 2790 3606 4763 6319 7807 8918 9311 13431 15723 19953 20935 21092 21286
9	1921 2131 2321 3114 4589 5133 5477 8265 9891 13941 14404 15777 17310 17787 18399 20916
10	1612 3502 3696 5084 6421 7410 7723 8467 8787 9434 11516 14329 14505 17723 19229 19308
11	2900 3311 3430 3984 4843 5422 6049 7374 7572 11037 15112 15173 17144 20378 20718 20854
12	4685 4896 7712 9120 10019 11988 12657 12907 13113 13519 14384 15347 16180 17125 19923 20661
13	189 2543 5548 6001 8979 9224 12641 13404 13505 13674 16011 16234 16820 17230 18945 19912
14	788 6540 7724 9898 13150 13817 15313 15554 15928 16118 17734 18170 18386 19422 20106 20931
15	4779 4964 6722 8474 9298 10620 11326 11471 12897 13482 15805 17076 18193 18260 20122 21392
16	385 2524 3486 4503 6708 7712 8632 8908 9283 10826 12081 12699 16551 16846 19058 20749
17	1405 1748 3058 4219 9053 9906 10581 11242 11515 14910 15143 16499 18395 18853 19454 21264
18	1875 2716 5358 6878 7089 8758 9659 11909 12290 12697 14631 17200 18305 18973 19159 21583
19	683 1564 1718 3350 3940 5672 6189 9361 11347 11915 13236 15946 16404 18372 21116 21282
20	962 4602 5035 5827 8007 10139 11524 11970 13479 13586 16061 18532 18705 19152 19625 20054
21	497 1698 1976 2383 2823 3479 7527 9948 11889 13649 14491 15431 16868 17232 18316 20453
22	1029 6199 8477 9707 10400 10913 11617 11923 12482 14690 14988 15796 16104 19272 20426 20731
23	95 953 1208 1818 5640 5797 8852 9399 9595 10877 11087 13129 15122 16631 17643 18867
24	12575 12680 15845
25	5870 6972 16463
26	2025 3655 7716
27	4258 6387 14477
28	12282 13274 18603
29	1657 10810 12509
30	839 8734 21409
31	4038 5993 15640
32	14005 15282 18931
33	4342 7977 17828
34	14144 16500 21426
35	592 4952 13747
36	8156 8859 18093
37	5995 7267 9133
38	5581 17111 17306
39	3218 7635 14267
40	5844 13026 18796
41	5686 12821 15609
42	3980 4228 7731
43	75 18922 20730
44	18712 20866 21302
45	10674 13003 20481
46	1914 6964 18238
47	12972 13411 21559
48	13207 16406 19548
49	4021 6039 6320
50	4721 20336 20819

TABLE 15-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
51	2797 15321 20509
52	8187 8774 19113
53	3634 10487 13963
54	3205 4325 12098
55	1444 4409 13667
56	1163 5856 7623
57	2324 10790 14560
58	1468 12791 17743
59	9765 12262 20117
60	1456 4150 11096
61	6900 9631 15217
62	4009 5995 7322
63	10673 11315 17310
64	5792 10504 18221
65	4748 14299 14994
66	4176 14868 20718
67	6147 15884 19749
68	9044 10345 20757
69	5817 7873 11969
70	3114 3924 8326
71	5535 6651 16116
72	7642 16391 17673
73	15798 17918 19172
74	1181 5291 17166
75	4220 4567 18197
76	1255 17730 19449
77	2099 5538 14774
78	6542 7475 17228
79	3418 4801 20715
80	1145 4245 16632
81	2034 11271 21000
82	7238 8108 15208
83	193 11374 15841
84	1333 5056 12441
85	8026 17906 18037
86	162 4432 8739
87	4762 5268 8940
88	3494 6384 17600
89	3029 6710 11442
90	919 9289 19407
91	9130 17762 20598
92	5140 17911 20878
93	14153 16376 19323
94	3863 4847 21567
95	2608 4840 14455
96	1117 4061 5255
97	13468 14536 14928
98	9068 13023 13346
99	1139 14402 15445
100	2190 17004 17906
101	2989 5524 12399
102	8489 8899 15486
103	6683 6970 13387
104	3745 9975 17753
105	1250 4246 17193
106	9992 19441 19965
107	5796 7986 21297
108	2781 3232 16020
109	6891 7654 15969
110	4002 13033 19217
111	7439 9192 13183
112	1390 8673 18485
113	24 4845 14633
114	6083 14765 15640
115	9652 13452 21404
116	7787 10616 17371
117	2959 6783 13581
118	11596 18575 20878
119	8318 14614 20870

As another example of results designed in consideration of these additional conditions, if a codeword length N_{ldpc} is 64800, a code rate R is 12/15, M is 360, and Q_{ldpc} is 36, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 16.

TABLE 16

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	584 1472 1621 1867 3338 3568 3723 4185 5126 5889 7737 8632 8940 9725
1	221 445 590 3779 3835 6939 7743 8280 8448 8491 9367 10042 11242 12917
2	4662 4837 4900 5029 6449 6687 6751 8684 9936 11681 11811 11886 12089 12909
3	2418 3018 3647 4210 4473 7447 7502 9490 10067 11092 11139 11256 12201 12383
4	2591 2947 3349 3406 4417 4519 5176 6672 8498 8863 9201 11294 11376 12184
5	27 101 197 290 871 1727 3911 5411 6676 8701 9350 10310 10798 12439
6	1765 1897 2923 3584 3901 4048 6963 7054 7132 9165 10184 10824 11278 12669
7	2183 3740 4808 5217 5660 6375 6787 8219 8466 9037 10353 10583 11118 12762
8	73 1594 2146 2715 3501 3572 3639 3725 6959 7187 8406 10120 10507 10691
9	240 732 1215 2185 2788 2830 3499 3881 4197 4991 6425 7061 9756 10491
10	831 1568 1828 3424 4319 4516 4639 6018 9702 10203 10417 11240 11518 12458
11	2024 2970 3048 3638 3676 4152 5284 5779 5926 9426 9945 10873 11787 11837
12	1049 1218 1651 2328 3493 4363 5750 6483 7613 8782 9738 9803 11744 11937
13	1193 2060 2289 2964 3478 4592 4756 6709 7162 8231 8326 11140 11908 12243
14	978 2120 2439 3338 3850 4589 6567 8745 9656 9708 10161 10542 10711 12639
15	2403 2938 3117 3247 3711 5593 5844 5932 7801 10152 10226 11498 12162 12941
16	1781 2229 2276 2533 3582 3951 5279 5774 7930 9824 10920 11038 12340 12440
17	289 384 1980 2230 3464 3873 5958 8656 8942 9006 10175 11425 11745 12530
18	155 354 1090 1330 2002 2236 3559 3705 4922 5958 6576 8564 9972 12760
19	303 876 2059 2142 5244 5330 6644 7576 8614 9598 10410 10718 11033 12957
20	3449 3617 4408 4602 4727 6182 8835 8928 9372 9644 10237 10747 11655 12747
21	811 2565 2820 8677 8974 9632 11069 11548 11839 12107 12411 12695 12812 12890
22	972 4123 4943 6385 6449 7339 7477 8379 9177 9359 10074 11709 12552 12831
23	842 973 1541 2262 2905 5276 6758 7099 7894 8128 8325 8663 8875 10050
24	474 791 968 3902 4924 4965 5085 5908 6109 6329 7931 9038 9401 10568
25	1397 4461 4658 5911 6037 7127 7318 8678 8924 9000 9473 9602 10446 12692
26	1334 7571 12881
27	1393 1447 7972
28	633 1257 10597
29	4843 5102 11056
30	3294 8015 10513
31	1108 10374 10546
32	5353 7824 10111
33	3398 7674 8569
34	7719 9478 10503
35	2997 9418 9581
36	5777 6519 11229
37	1966 5214 9899
38	6 4088 5827
39	836 9248 9612
40	483 7229 7548
41	7865 8289 9804
42	2915 11098 11900
43	6180 7096 9481
44	1431 6786 8924
45	748 6757 8625
46	3312 4475 7204
47	1852 8958 11020
48	1915 2903 4006
49	6776 10886 12531
50	2594 9998 12742
51	159 2002 12079
52	853 3281 3762
53	5201 5798 6413
54	3882 6062 12047
55	4133 6775 9657
56	228 6874 11183
57	7433 10728 10864
58	7735 8073 12734
59	2844 4621 11779
60	3909 7103 12804
61	6002 9704 11060
62	5864 6856 7681
63	3652 5869 7605
64	2546 2657 4461
65	2423 4203 9111
66	244 1855 4691
67	1106 2178 6371

TABLE 16-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
68	391 1617 10126
69	250 9259 10603
70	3435 4614 6924
71	1742 8045 9529
72	7667 8875 11451
73	4023 6108 6911
74	8621 10184 11650
75	6726 10861 12348
76	3228 6302 7388
77	1 1137 5358
78	381 2424 8537
79	3256 7508 10044
80	1980 2219 4569
81	2468 5699 10319
82	2803 3314 12808
83	8578 9642 11533
84	829 4585 7923
85	59 329 5575
86	1067 5709 6867
87	1175 4744 12219
88	109 2518 6756
89	2105 10626 11153
90	5192 10696 10749
91	6260 7641 8233
92	2998 3094 11214
93	3398 6466 11494
94	6574 10448 12160
95	2734 10755 12780
96	1028 7958 10825
97	8545 8602 10793
98	392 3398 11417
99	6639 9291 12571
100	1067 7919 8934
101	1064 2848 12753
102	6076 8656 12690
103	5504 6193 10171
104	1951 7156 7356
105	4389 4780 7889
106	526 4804 9141
107	1238 3648 10464
108	2587 5624 12557
109	5560 5903 11963
110	1134 2570 3297
111	10041 11583 12157
112	1263 9585 12912
113	3744 7898 10646
114	45 9074 10315
115	1051 6188 10038
116	2242 8394 12712
117	3598 9025 12651
118	2295 3540 5610
119	1914 4378 12423
120	1766 3635 12759
121	5177 9586 11143
122	943 3590 11649
123	4864 6905 10454
124	5852 6042 10421
125	6095 8285 12349
126	2070 7171 8563
127	718 12234 12716
128	512 10667 11353
129	3629 6485 7040
130	2880 8865 11466
131	4490 10220 11796
132	5440 8819 9103
133	5262 7543 12411
134	516 7779 10940
135	2515 5843 9202
136	4684 5994 10586
137	573 2270 3324
138	7870 8317 10322
139	6856 7638 12909
140	1583 7669 10781
141	8141 9085 12555
142	3903 5485 9992
143	4467 11998 12904

As another example of results designed in consideration of these additional conditions, if a codeword length N_{ldpc} is 64800, a code rate R is $6/15$, M is 360, and Q_{ldpc} is 108, indices of a row at which one (1) is positioned in a zero (0)-th column of an i -th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 17.

TABLE 17

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	1606 3402 4961 6751 7132 11516 12300 12482 12592 13342 13764 14123 21576 23946 24533 25376 25667 26836 31799 34173 35462 36153 36740 37085 37152 37468 37658
1	4621 5007 6910 8732 9757 11508 13099 15513 16335 18052 19512 21319 23663 25628 27208 31333 32219 33003 33239 33447 36200 36473 36938 37201 37283 37495 38642
2	16 1094 2020 3080 4194 5098 5631 6877 7889 8237 9804 10067 11017 11366 13136 13354 15379 18934 20199 24522 26172 28666 30386 32714 36390 37015 37162
3	700 897 1708 6017 6490 7372 7825 9546 10398 16605 18561 18745 21625 22137 23693 24340 24966 25015 26995 28586 28895 29687 33938 34520 34858 37056 38297
4	159 2010 2573 3617 4452 4958 5556 5832 6481 8227 9924 10836 14954 15594 16623 18065 19249 22394 22677 23408 23731 24076 24776 27007 28222 30343 38371
5	3118 3545 4768 4992 5227 6732 8170 9397 10522 11508 15536 20218 21921 28599 29445 29758 29968 31014 32027 33685 34378 35867 36323 36728 36870 38335 38623
6	1264 4254 6936 9165 9486 9950 10861 11653 13697 13961 15164 15665 18444 19470 20313 21189 24371 26431 26999 28086 28251 29261 31981 34015 35850 36129 37186
7	111 1307 1628 2041 2524 5358 7988 8191 10322 11905 12919 14127 15515 15711 17061 19024 21195 22902 23727 24401 24608 25111 25228 27338 35398 37794 38196
8	961 3035 7174 7948 13355 13607 14971 18189 18339 18665 18875 19142 20615 21136 21309 21758 23366 24745 25849 25982 27583 30006 31118 32106 36469 36583 37920
9	2990 3549 4273 4808 5707 6021 6509 7456 8240 10044 12262 12660 13085 14750 15680 16049 21587 23997 25803 28343 28693 34393 34860 35490 36021 37737 38296
10	955 4323 5145 6885 8123 9730 11840 12216 19194 20313 23056 24248 24830 25268 26617 26801 28557 29753 30745 31450 31973 32839 33025 33296 35710 37366 37509
11	264 605 4181 4483 5156 7238 8863 10939 11251 12964 16254 17511 20017 22395 22818 23261 23422 24064 26329 27723 28186 30434 31956 33971 34372 36764 38123
12	520 2562 2794 3528 3860 4402 5676 6963 8655 9018 9783 11933 16336 17193 17320 19035 20606 23579 23769 24123 24966 27866 32457 34011 34499 36620 37526
13	10106 10637 10906 34242
14	1856 15100 19378 21848
15	943 11191 27806 29411
16	4575 6359 13629 19383
17	4476 4953 18782 24313
18	5441 6381 21840 35943
19	9638 9763 12546 30120
20	9587 10626 11047 25700
21	4088 15298 28768 35047
22	2332 6363 8782 28863
23	4625 4933 28298 30289
24	3541 4918 18257 31746
25	1221 25233 26757 34892
26	8150 16677 27934 30021
27	8500 25016 33043 38070
28	7374 10207 16189 35811
29	611 18480 20064 38261
30	25416 27352 36089 38469
31	1667 17614 25839 32776
32	4118 12481 21912 37945
33	5573 13222 23619 31271
34	18271 26251 27182 30587
35	14690 26430 26799 34355
36	13688 16040 20716 34558
37	2740 14957 23436 32540
38	3491 14365 14681 36858
39	4796 6238 25203 27854
40	1731 12816 17344 26025
41	19182 21662 23742 27872
42	6502 13641 17509 34713
43	12246 12372 16746 27452
44	1589 21528 30621 34003
45	12328 20515 30651 31432
46	3415 22656 23427 36395
47	632 5209 25958 31085
48	619 3690 19648 37778
49	9528 13581 26965 36447
50	2147 26249 26968 28776
51	15698 18209 30683
52	1132 19888 34111
53	4608 25513 38874
54	475 1729 34100
55	7348 32277 38587

TABLE 17-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
56	182 16473 33082
57	3865 9678 21265
58	4447 20151 27618
59	6335 14371 38711
60	704 9695 28858
61	4856 9757 30546
62	1993 19361 30732
63	756 28000 29138
64	3821 24076 31813
65	4611 12326 32291
66	7628 21515 34995
67	1246 13294 30068
68	6466 33233 35865
69	14484 23274 38150
70	21269 36411 37450
71	23129 26195 37653

As another example of results designed in consideration of these additional conditions, if a codeword length N_{ldpc} is 64800, a code rate R is 10/15, M is 360, and Q_{ldpc} is 60, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group of a parity-check matrix having the structure of FIG. 3 are as shown in following Table 18.

TABLE 18

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
0	979 1423 4166 4609 6341 8258 10334 10548 14098 14514 17051 17333 17653 17830 17990
1	2559 4025 6344 6510 9167 9728 11312 14856 17104 17721 18600 18791 19079 19697 19840
2	3243 6894 7950 10539 12042 13233 13938 14752 16449 16727 17025 18297 18796 19400 21577
3	3272 3574 6341 6722 9191 10807 10957 12531 14036 15580 16651 17007 17309 19415 19845
4	155 4598 10201 10975 11086 11296 12713 15364 15978 16395 17542 18164 18451 18612 20617
5	1128 1999 3926 4069 5558 6085 6337 8386 10693 12450 15438 16223 16370 17308 18634
6	2408 2929 3630 4357 5852 7329 8536 8695 10603 11003 14304 14937 15767 18402 21502
7	199 3066 6446 6849 8973 9536 10452 12857 13675 15913 16717 17654 19802 20115 21579
8	312 870 2095 2586 5517 6196 6757 7311 7368 13046 15384 18576 20349 21424 21587
9	985 1591 3248 3509 3706 3847 6174 6276 7864 9033 13618 15675 16446 18355 18843
10	975 3774 4083 5825 6166 7218 7633 9657 10103 13052 14240 17320 18126 19544 20208
11	1795 2005 2544 3418 6148 8051 9066 9725 10676 10752 11512 15171 17523 20481 21059
12	167 315 1824 2325 2640 2868 6070 6597 7016 8109 9815 11608 16142 17912 19625
13	1298 1896 3039 4303 4690 8787 12241 13600 14478 15492 16602 17115 17913 19466 20597
14	568 3695 6045 6624 8131 8404 8590 9059 9246 11570 14336 18657 18941 19218 21506
15	228 1889 1967 2299 3011 5074 7044 7596 7689 9534 10244 10697 11691 17902 21410
16	1330 1579 1739 2234 3701 3865 5713 6677 7263 11172 12143 12765 17121 20011 21436
17	303 1668 2501 4925 5778 5985 9635 10140 10820 11779 11849 12058 15650 20426 20527
18	698 2484 3071 3219 4054 4125 5663 5939 6928 7086 8054 12173 16280 17945 19302
19	232 1619 3040 4901 7438 8135 9117 9233 10131 13321 17347 17436 18193 18586 19929
20	12 3721 6254 6609 7880 8139 10437 12262 13928 14065 14149 15032 15694 16264 18883
21	482 915 1548 1637 6687 9338 10163 11768 11970 15524 15695 17386 18787 19210 19340

TABLE 18-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group										
22	1291	2500	4109	4511	5099	5194	10014	13165	13256	13972	15409
	18584	20998									
23	1761	4778	7444	7740	8129	8341	8931	9136	9207	10003	10678
	20990										
24	3060	3522	5361	5692	6833	8342	8792	11023	11211	11548	11914
	15541	19707									
25	1322	2348	2970	5632	6349	7577	8782	9113	9267	9376	12042
	21321										
26	6785	11960	21455								
27	1223	15672	19550								
28	5976	11335	20385								
29	2818	9387	15317								
30	2763	3554	18102								
31	5230	11489	18997								
32	5809	15779	20674								
33	2620	17838	18533								
34	3025	9342	9931								
35	3728	5337	12142								
36	2520	6666	9164								
37	12892	15307	20912								
38	10736	12393	16539								
39	1075	2407	12853								
40	4921	5411	18206								
41	5955	15647	16838								
42	6384	10336	19266								
43	429	10421	17266								
44	4880	10431	12208								
45	2910	11895	12442								
46	7366	18362	18772								
47	4341	7903	14994								
48	4564	6714	7378								
49	4639	8652	18871								
50	15787	18048	20246								
51	3241	11079	13640								
52	1559	2936	15881								
53	2737	6349	10881								
54	10394	16107	17073								
55	8207	9043	12874								
56	7805	16058	17905								
57	11189	15767	17764								
58	5823	12923	14316								
59	11080	20390	20924								
60	568	8263	17411								
61	1845	3557	6562								
62	2890	10936	14756								
63	9031	14220	21517								
64	3529	12955	15902								
65	413	6750	8735								
66	6784	12092	16421								
67	12019	13794	15308								
68	12588	15378	17676								
69	8067	14589	19304								
70	1244	5877	6085								
71	15897	19349	19993								
72	1426	2394	12264								
73	3456	8931	12075								
74	13342	15273	20351								
75	9138	13352	20798								
76	7031	7626	14081								
77	4280	4507	15617								
78	4170	10569	14335								
79	3839	7514	16578								
80	4688	12815	18782								
81	4861	7858	9435								
82	605	5445	12912								
83	2280	4734	7311								
84	6668	8128	12638								
85	3733	10621	19534								
86	13933	18316	19341								
87	1786	3037	21566								
88	2202	13239	16432								
89	4882	5808	9300								
90	4580	8484	16754								
91	14630	17502	18269								
92	6889	11119	12447								
93	8162	9078	16330								
94	6538	17851	18100								
95	17763	19793	20816								

TABLE 18-continued

i	Indices of rows at which 1s are positioned in 0-th column of i-th column group
96	2183 11907 17567
97	6640 14428 15175
98	877 12035 14081
99	1336 6468 12328
100	5948 9146 12003
101	3782 5699 12445
102	1770 7946 8244
103	7384 12639 14989
104	1469 11586 20959
105	7943 10450 15907
106	5005 8153 10035
107	17750 18826 21513
108	4725 8041 10112
109	3837 16266 17376
110	11340 17361 17512
111	1269 4611 4774
112	2322 10813 16157
113	16752 16843 18959
114	70 4325 18753
115	3165 8153 15384
116	160 8045 16823
117	14112 16724 16792
118	4291 7667 18176
119	5943 19879 20721

Here, it is to be noted that even if the parity-check matrix in which an order of indices within a sequence for each i-th column group in any of above Tables 9 to 18 is changed, the changed parity-check matrix may also be applied to the same codes to which the original parity-check matrix applies.

For example, as shown in above Table 11, the sequence corresponding to the zero (0)-th column group is arranged in an order of 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140, and even if the sequence has a changed index order like 2135, 8943, 4635, 114, 3045, 10109, 13161, 21140, 5681, 6571, 5512, 19328, 14218, 13668, 17417, 10053, a parity-check matrix having the changed sequence may be used for the same codes.

In addition, even if one index sequence of one column group is changed to an index sequence of another column group and vice versa, that is, an index sequence of the other column group is changed to the index sequence of the first column group, in above Tables 9 to 18, algebraic characteristics such as the cycle characteristics and the degree distributions on a graph of codes are not changed. Therefore, this case of changing index sequences between column groups of a parity-check matrix is another exemplary embodiment of the inventive concept.

For example, in above Table 11, the index sequence 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140 of the zero (0)-th column group and the index sequence 19, 768, 1263, 3305, 6513, 7677, 7956, 9040, 13427, 16641, 17280, 18452, 18584, 18925, 19559, 20587 of a twelfth column group can be changed to set the sequence of the zero (0)-th column group to be 19, 768, 1263, 3305, 6513, 7677, 7956, 9040, 13427, 16641, 17280, 18452, 18584, 18925, 19559, 20587, and the sequence of the twelfth column group is set to be 114, 2135, 3045, 4635, 5512, 5681, 6571, 8943, 10053, 10109, 13161, 13668, 14218, 17417, 19328, 21140. That is, even if the index sequences are exchanged between the zero (0)-th column group and the twelfth column group, the cycle characteristics, the degree distributions, and the like in terms of the graph of codes are not changed (actually, since exchanging the index sequences between corresponding

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column groups is the same as changing only the arranging order of column groups within a parity-check matrix, the main algebraic characteristics are not changed).

In addition, a parity-check matrix in which an integer multiple of Q_{ldpc} is added to all indices of any column group in above Tables 9 to 18 may also result in the same algebraic characteristics such as the cycle characteristics and the degree distributions on the graph of the codes, according to an exemplary embodiment.

For example, if a multiple of 60 (that is, $Q_{ldpc}=(N_{ldpc}-K_{ldpc})/M=60$) is added to all indices 12575, 15845 and 18200 of a twenty-fourth column group in above Table 11, that is, the indices are changed to 12635 ($=12575+60$), 15905 ($=15845+60$), and 18260 ($=18200+60$), the algebraic characteristics such as the cycle characteristics and the degree distributions on the graph of codes are not changed (actually, since a sequence obtained by adding an integer multiple of Q_{ldpc} in a column group has the same effect as rearranging only the order of columns within the column group, the main algebraic characteristics are not changed).

Here, it is to be noted that in the case in which an index value obtained by adding an integer multiple of Q_{ldpc} to a given index sequence is a value of $N_{ldpc}-K_{ldpc}$ or more, the index value is changed and applied to a value obtained by performing a modulo operation on $N_{ldpc}-K_{ldpc}$.

For example, since a result of adding only 60×60 to the sequence 12575, 15845 and 18200 of the twenty-fourth column group in above Table 11 becomes 16175, 19445 and 21800 while $N_{ldpc}-K_{ldpc}=21600$, the changed sequence may be 16175, 19445 and 200 or 200, 16175 and 19445 by applying modulo-21600 to a result of adding only the integer multiple of Q_{ldpc} to the given sequence.

Hereinafter, a process of encoding LDPC codes using a parity-check matrix having the structure as illustrated in FIG. 3 will be described. As described above, the process of encoding LDPC codes is to determine a codeword C meeting a relational equation: parity-check matrix $\times$ codeword = 0. That is, the process of encoding LDPC codes may be represented by $H \cdot C^T = 0$. Here, H is the parity-check matrix and C represents the LDPC codeword.

Hereinafter, if it is assumed that LDPC encoded information word bits are $(i_0, i_1, \dots, i_{K_{ldpc}-1})$ and the LDPC

codeword bits generated by the LDPC encoding are $(c_0, c_1, \dots, c_{N_{ldpc}-1})$, a method for calculating LDPC codeword bits will be described.

First, since the LDPC code is a systematic code, c_k for $0 < k < K_{ldpc} - 1$ is set to be the same as i_k . In addition, the remaining codeword bits are set to be $p_k := c_{k+K_{ldpc}}$. Here, p_k is parity bits and may be calculated as described below.

Meanwhile, according to an exemplary embodiment, since the parity-check matrix is defined by in above Tables 9 to 18, a process to be described below may be applied in the case in which the parity-check matrix is defined by in above Tables 9 to 18.

First, if it is assumed that an entry notated in a j -th position of an i -th row in above Tables 9 to 18 is $q(i, j, l) = q(i, j, 0) + Q_{ldpc} \cdot l \pmod{N_{ldpc} - K_{ldpc}}$ for $0 < l < 360$. Here, accumulation '+' means additions defined in a Galois field (GF) (2) (that is, additions in GF (2)). In addition, Q_{ldpc} , which is a size of each column cyclically shifted in an information word sub-matrix, may be a value defined in above Tables 9 to 18, respectively.

Meanwhile, when $q(i, j, 0)$ and $q(i, j, 1)$ are defined as described above, a process of calculating parity bits is as follows.

Step 1) The parity bits are initialized to '0'. That is, $p_k = 0$ for $0 < k < N_{ldpc} - K_{ldpc}$.

Step 2) i and l are set to be $i = \lfloor k/360 \rfloor$ and $l = k \pmod{360}$ for all k values of $0 < k < K_{ldpc}$. Here, $\lfloor x \rfloor$ is the largest integer value among integers that are not larger than x . That is, $\lfloor 1.2 \rfloor = 1$. Next, i_k is added to $p_{q(i, j, l)}$ for all j s as following based on the set i and l values. That is, $p_{q(i, 0, l)} = p_{q(i, 0, l)} + i_k$, $p_{q(i, 1, l)} = p_{q(i, 1, l)} + i_k$, $p_{q(i, 2, l)} = p_{q(i, 2, l)} + i_k$, $\dots$, $p_{q(i, w(i)-1, l)} = p_{q(i, w(i)-1, l)} + i_k$ are calculated.

Here, $w(i)$ is the number of values of the i -th row in above Tables 9 to 18 and represents the number of ones (1s) in a column corresponding to i_k in the parity-check matrix. In addition, $q(i, j, 0)$, which is the entry notated in the j -th position of the i -th row in above Tables 9 to 18, is an index of the parity bit and represents a position of a row at which one (1) is positioned in a column corresponding to i_k in the parity-check matrix.

Step 3) $p_k = p_k + p_{k-1}$ is calculated for all k s meeting $0 < k < N_{ldpc} - K_{ldpc}$ to calculate the parity bits p_k .

The parity bits are calculated by the foregoing method, such that all the LDPC codeword bits $c_0, c_1, \dots, c_{N_{ldpc}-1}$ may be calculated.

Meanwhile, the LDPC encoding process as described above is only an example. That is, since the LDPC encoding process is a process of calculating an LDPC codeword C satisfying $H \cdot C^T = 0$, various encoding methods for the given parity-check matrix may be present.

For example, a scheme applied in the DVB-T2 standard may also be applied to the case in which the parity-check matrix is defined by above Tables 9 to 18. Hereinafter, the LDPC encoding process according to the scheme described in the DVB-T2 standard will be schematically described using an example in which the parity-check matrix is defined in above Table 11.

First, if it is assumed that information word bits having a length of K_{ldpc} are $[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}]$ and parity bits having a length of $N_{ldpc} - K_{ldpc}$ are $[p_0, p_1, p_2, \dots, p_{N_{ldpc}-K_{ldpc}-1}]$, the LDPC encoding may be performed by the following process.

Step 1) The parity bits are initialized to '0'. That is, $p_0 = p_1 = p_2 = \dots = p_{N_{ldpc}-K_{ldpc}-1}$.

Step 2) A zero (0)-th information word bit i_0 is accumulated in a parity bit which has an address of a parity bit defined in a first row (that is, a row of $i=0$) of above

Table 11 as the index of the parity bit. This may be represented by following mathematical expressions 11.

$$\begin{aligned} p_{114} &= p_{114} \oplus i_0 & p_{10053} &= p_{10053} \oplus i_0 \\ p_{2135} &= p_{2135} \oplus i_0 & p_{10109} &= p_{10109} \oplus i_0 \\ p_{3045} &= p_{3045} \oplus i_0 & p_{13161} &= p_{13161} \oplus i_0 \\ p_{4635} &= p_{4635} \oplus i_0 & p_{13668} &= p_{13668} \oplus i_0 \\ p_{5512} &= p_{5512} \oplus i_0 & p_{14218} &= p_{14218} \oplus i_0 \\ p_{5681} &= p_{5681} \oplus i_0 & p_{17417} &= p_{17417} \oplus i_0 \\ p_{6571} &= p_{6571} \oplus i_0 & p_{19328} &= p_{19328} \oplus i_0 \\ p_{8943} &= p_{8943} \oplus i_0 & p_{21140} &= p_{21140} \oplus i_0 \end{aligned} \quad (11)$$

In the above mathematical expressions, i_0 represents a zero (0)-th information word bit, p_i represents an i -th parity bit, and $\oplus$ represents a binary operation. According to the binary operation, $1 \oplus 1$ is $1 \oplus 0$ is 1, $0 \oplus 1$ is 1, and $0 \oplus 0$ is 0.

Step 3) The remaining 359 information word bits i_m ($m=1, 2, \dots, 359$) are accumulated in the parity bits. Here, the remaining information word bits may be information word bits which belong to the same column group as a column group to which i_0 belongs. In this case, the addresses of the parity bits may be determined based on following mathematical expression 12.

$$(x + (m \pmod{360}) \times Q_{ldpc}) \pmod{(N_{ldpc} - K_{ldpc})} \quad (12)$$

In above mathematical expression 12, x is an address of a parity bit accumulator corresponding to the information word bit i_0 , and Q_{ldpc} is 60 as a size of each column which is shifted in a sub-matrix corresponding to the information word.

As a result, each of the information word bits i_m ($m=1, 2, \dots, 359$) is accumulated in each parity bit having the addresses of the parity bits calculated based on above mathematical expression 12 as indices. As an example, operations as represented by following mathematical expression 13s may be performed on the information word bit

$$\begin{aligned} p_{174} &= p_{174} \oplus i_1 & p_{10113} &= p_{10113} \oplus i_1 \\ p_{2195} &= p_{2195} \oplus i_1 & p_{10169} &= p_{10169} \oplus i_1 \\ p_{3105} &= p_{3105} \oplus i_1 & p_{13221} &= p_{13221} \oplus i_1 \\ p_{4695} &= p_{4695} \oplus i_1 & p_{13728} &= p_{13728} \oplus i_1 \\ p_{5572} &= p_{5572} \oplus i_1 & p_{14278} &= p_{14278} \oplus i_1 \\ p_{5741} &= p_{5741} \oplus i_1 & p_{17477} &= p_{17477} \oplus i_1 \\ p_{6631} &= p_{6631} \oplus i_1 & p_{19388} &= p_{19388} \oplus i_1 \\ p_{9003} &= p_{9003} \oplus i_1 & p_{21200} &= p_{21200} \oplus i_1 \end{aligned} \quad (13)$$

In above mathematical expression, i_1 represents a 1-th information word bit, p_i represents an i -th parity bit, and $\oplus$ represents a binary operation. According to the binary operation, $1 \oplus 1$ is 0, $1 \oplus 0$ is 1, $0 \oplus 1$ is 1, and $0 \oplus 0$ is 0.

Step 4) A 360-th information word bit i_{360} is accumulated in a parity bit which has an address of a parity bit defined in a second row (that is, a row of $i=1$) of above Table 11 as the index of the parity bit.

Step 5) The remaining 359 information word bits belonging to the same group as a group to which the infor-

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mation word bit i_{360} belongs are accumulated in the parity bit. In this case, the address of the parity bit may be determined based on above mathematical expression 6. However, in this case, x is an address of a parity bit accumulator corresponding to the information word bit i_{360} .

Step 6) the foregoing processes of Step 4 and Step 5 are repeated for all of the column groups of above Table 5.

Step 7) As a result, the parity bit p_i is calculated based on following mathematical expression 14. Here, i is initialized to one (1).

$$p_i = p_i \oplus p_{i-1} \quad i=1, 2, \dots, N_{ldpc} - K_{ldpc} - 1 \quad (14)$$

In above mathematical expression 14, p_i represents an i -th parity bit, N_{ldpc} represents the LDPC codeword length, K_{ldpc} represents the information word length in the LDPC codeword, and $\oplus$ represents a binary operation. According to the binary operation, $1 \oplus 1$ is 0, $1 \oplus 0$ is 1, $0 \oplus 1$ is 1, and $0 \oplus 0$ is 0.

According to the above method, the parity bits may be calculated.

Meanwhile, the addresses of the parity bit present in the zero (0)-th column of the i -th column group are the same as the indices of the row at which one (1) is positioned in the zero (0)-th column of the i -th column group. Therefore, the indices of the row at which one (1) is positioned in the zero (0)-th column of the i -th column group in above Tables 9 to 18 are represented as addresses of the parity bits in the encoding process. Therefore, above Tables 9 to 18 may show "addresses of parity bit accumulators".

As described above, according to the above exemplary embodiments, the LDPC encoding process may be performed using various schemes to generate an LDPC codeword.

Meanwhile, LDPC codes may be decoded using an iterative decoding algorithm based on the sum-product algorithm on the bipartite graph illustrated in FIG. 2, in which the sum-product algorithm is a kind of message passing algorithm.

Hereinafter, a message passing operation generally used for LDPC decoding will be described with reference to FIGS. 5A and 5B.

FIGS. 5A and 5B illustrate a message passing operation in any check node and variable node for LDPC decoding, according to an exemplary embodiment.

FIG. 5A illustrates a check node m 500 and a plurality of variable nodes 510, 520, 530 and 540 connected to the check node m 500. In addition, $T_{n', m}$ illustrated in FIG. 5A indicates a message passing from the variable node n' 510 to the check node m 500, and $E_{n, m}$ indicates a message passing from the check node m 500 to the variable node n 530. Here, a set of all variable nodes connected to the check node m 500 is defined by $N(m)$, and a set except the variable node n 530 in the $N(m)$ is defined by $N(m) \setminus n$.

In this case, a message update rule based on the sum-product algorithm may be represented by following mathematical expressions 15.

$$|E_{n, m}| = \Phi \left[\sum_{n' \in N(m) \setminus n} \Phi(|T_{n', m}|) \right] \quad (15)$$

$$\text{Sign}(E_{n, m}) = \prod_{n' \in N(m) \setminus n} \text{sign}(T_{n', m})$$

In above mathematical expressions 15, $\text{Sign}(E_{n, m})$ indicates a sign of the message $E_{n, m}$, and $E_{n, m}$ indicates a

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magnitude of the message $E_{n, m}$. Meanwhile, a function $\Phi(x)$ may be represented by following mathematical expression 16.

$$\Phi(x) = -\log \left(\tanh \left(\frac{x}{2} \right) \right) \quad (16)$$

FIG. 5B illustrates a variable node x 550 and a plurality of check nodes 560, 570, 580 and 590 connected to the variable node x 550. In addition, $E_{y', x}$ illustrated in FIG. 5B indicates a message passing from the check node y' 560 to the variable node x 550, and $T_{y, x}$ indicates a message passing from the variable node x 550 to the variable node y 580. Here, a set of all variable nodes connected to the variable node x 550 is defined by $M(x)$, and a set except the check node y 580 in $M(x)$ is defined by $M(x) \setminus y$. In this case, the message update rule based on the sum-product algorithm may be represented by following mathematical expression 17.

$$T_{y, x} = E_x + \sum_{y' \in M(x) \setminus y} E_{y', x} \quad (17)$$

In above mathematical expression 17, E_x represents an initial message value of the variable node x .

In addition, when a bit value of the node x is decided, it may be represented by following mathematical expression 18.

$$P_x = E_x + \sum_{y' \in M(x)} E_{y', x} \quad (18)$$

In this case, an encoding bit corresponding to the node x may be decided depending on a value of P_x .

In FIG. 5, the foregoing method is a general decoding method, and thus, the detailed description thereof will be omitted. However, in addition to the method described with reference to FIG. 5, other methods for determining the passing message values at the variable node and the check node may also be applied (Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger, "Factor Graphs and the Sum-Product Algorithm," IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. 47, NO. 2, FEBRUARY 2001, pp 498-519).

FIG. 6 is a block diagram illustrating a configuration of an encoding apparatus according to an exemplary embodiment. The encoding apparatus 600 may perform the LDPC encoding described above.

As illustrated in FIG. 6, the encoding apparatus 600 includes an LDPC encoder 610. The LDPC encoder 610 may perform the LDPC encoding on input bits based on the parity-check matrix generated in the above method to generate an LDPC codeword. The LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

Here, the parity-check matrix may have the same structure as that of the parity-check matrix 300 illustrated in FIG. 3.

In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix described above.

Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M -th column. Here, M , which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

In this case, the LDPC encoder 610 may perform the LDPC encoding using parity-check matrices differently defined depending on a code rate (that is, a code rate of the LDPC codes).

For example, the LDPC encoder **610** may perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15, and perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC encoder **610** may perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15, and perform the LDPC encoding using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

Meanwhile, a detailed method for performing the LDPC encoding is already described, and thus, duplicate descriptions thereof will be omitted.

The encoding apparatus **600** may further include a memory (not illustrated) in which information on a code rate, a codeword length and a parity-check matrix of an LDPC code is previously stored, and the LDPC encoder **610** may perform LDPC encoding using this information. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

FIG. 7 is a block diagram showing a configuration of a transmitting apparatus according to an exemplary embodiment. As illustrated in FIG. 7, a transmitting apparatus **700** may include a Bose, Chaudhuri, Hocquenghem (BCH) encoder **710**, an LDPC encoder **720**, an interleaver **730**, and a modulator **740**.

The BCH encoder **710** performs BCH encoding on input bits and outputs a BCH codeword generated by BCH encoding to the LDPC encoder **720**.

In detail, the BCH encoder **710** performs the BCH encoding on the input bits $L=[i_0, i_1, \dots, i_{K_{bch}-1}]$, to generate $N_{ldpc}-K_{bch}$ BCH parity bits and generate an BCH codeword $l_{ldpc}=[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}]$. The BCH codeword $l_{ldpc}=[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}]$ is an information word for LDPC encoding, to the LDPC encoder **720**.

The BCH encoding is a well-known technology and is disclosed in "Bose, R. C.; Ray-Chaudhuri, D. K. (March 1960), "On A Class of Error Correcting Binary Group Codes", Information and Control 3 (1): 68-79, ISSN 0890-5401", etc., and thus, detailed descriptions thereof will be omitted herein.

Meanwhile, it may be changed whether the BCH encoder **710** is used. That is, in some cases, the BCH encoder **710** may also be omitted.

The LDPC encoder **720** performs LDPC encoding on the BCH codeword output from the BCH encoder **710** and outputs an LDPC codeword generated by the LDPC encoding to the interleaver **730**.

In detail, the LDPC encoder **720** performs the LDPC encoding on the BCH codeword $l_{ldpc}=[i_0, i_1, i_2, \dots, i_{K_{ldpc}-1}]$ output from the BCH encoder **710** as the information word to generate $N_{ldpc}-K_{ldpc}$ LDPC parity bits and generate the LDPC codeword $c_{ldpc}=[c_0, c_1, \dots, c_{N_{ldpc}-1}]$.

However, when the BCH encoder **710** is omitted, the LDPC encoder **720** may perform the LDPC encoding on the input bits.

Meanwhile, the LDPC encoder **720** of FIG. 7 may be implemented as the LDPC encoder **610** described with reference to FIG. 6. That is, the LDPC encoder **720** may perform the LDPC encoding using the parity-check matrix in which the information word sub-matrix is defined by above Tables 9 to 18 depending on the code rate and the parity sub-matrix has the dual diagonal structure.

To this end, the transmitting apparatus **700** may include a memory (not illustrated) for storing the information on the parity-check matrix. In this case, the parity-check matrix may be various depending on the code rate and may be the table defined by above Tables 9 to 18 as an example. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

The interleaver **730** performs interleaving on the LDPC codeword output from the LDPC encoder **720** and outputs the interleaved bits to the modulator **740**.

In this case, the interleaver **730** receives an LDPC codeword bit string output from the LDPC encoder **720** to perform the interleaving using a predetermined scheme. The interleaving scheme may be variously present and it may be variable whether or not the interleaver **730** is used.

The modulator **740** modulates the bit string output from the interleaver **730** and transmits the modulated bit string to a receiving apparatus (for example, **1000** of FIG. 10).

In detail, the modulator **740** may demultiplex bits output from the interleaver **730** and map the demultiplexed bits to constellation.

That is, the modulator **740** may convert the bits output from the interleaver **730** in a serial-to-parallel scheme to generate cells formed of a predetermined number of bits. Here, the number of bits forming each cell may be equal to the number of bits forming modulation symbols which are mapped to the constellation.

Next, the modulator **740** may map the demultiplexed bits to the constellation. That is, the modulator **740** modulates the demultiplexed bits using various modulation schemes such as QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, and 4096-QAM to generate the modulation symbols, and map the generated modulation symbols to constellation points. In this case, the demultiplexed bits are formed of the cells corresponding to the modulation symbols, and thus, each cell may sequentially be mapped to the constellation points.

Further, the modulator **740** may modulate signals mapped to the constellation and transmit the modulated signals to the receiving apparatus **1000**. For example, the modulator **740** may map the signals, which are mapped to the constellation, to an OFDM frame by using an orthogonal frequency division multiplexing (OFDM) scheme and transmit the mapped signals to the receiving apparatus **1000** through an allocated channel.

The transmitting apparatus **700** may pre-store various parameters which are used for encoding, interleaving, and modulation. Here, the parameters used for encoding may be information on the code rate and the codeword length of the BCH codes and the information on the code rate, the codeword length, and the parity-check matrix of the LDPC codes. Further, the parameters used for interleaving may be information on an interleaving rule and the parameter used for modulation may be information on the modulation schemes. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

In this case, each component configuring the transmitting apparatus **700** may be operated using the parameters.

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Meanwhile, although not illustrated, the transmitting apparatus **700** may further include a controller (not illustrated) for controlling an operation of the transmitting apparatus **700**.

In this case, the controller (not illustrated) may provide the information on the code rate and the codeword length of the BCH codes to the BCH encoder **710** and provide the information on the code rate, the codeword length, the parity-check matrix of the LDPC codes to the LDPC encoder **720**. Further, the controller (not illustrated) may provide the information on the interleaving schemes to the interleaver **730** and the information on the modulation schemes to the modulator **740**. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

FIG. **8** is a block diagram illustrating a configuration of a decoding apparatus according to an exemplary embodiment. As illustrated in FIG. **8**, a decoding apparatus **800** includes an LDPC decoder **810**.

The LDPC decoder **810** performs LDPC decoding on an LDPC codeword based on a parity-check matrix. Here, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

For example, the LDPC decoder **810** passes log likelihood ratio (LLR) values corresponding to the LDPC codeword bits using an iterative decoding algorithm to perform the LDPC decoding, thereby generating information word bits.

Here, the LLR values may be represented by channel values corresponding to the LDPC codeword bits by various methods.

For example, the LLR values may be represented by values obtained by taking a log on a ratio of a probability that bits transmitted through the channel at the transmitting side are zero (0) and a probability that bits are one (1). Further, the LLR values may be bit values determined by a soft decision and may also be a representative value determined depending on a section to which the probability that the bits transmitted from the transmitting side are zero (0) or one (1) belongs.

In this case, the transmitting side may use the LDPC encoder **610** as illustrated in FIG. **6** to generate the LDPC codeword.

Meanwhile, the parity-check matrix used for the LDPC decoding may have the same form as that of the parity-check matrix **300** illustrated in FIG. **3**.

In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix.

Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and is defined by the table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

In this case, the LDPC decoder **810** may perform the LDPC decoding using the parity-check matrices differently defined depending on a code rate (that is, a code rate of LDPC codes).

For example, the LDPC decoder **810** may perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15, and perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC decoder **810**

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may perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and perform the LDPC decoding using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

As described above, the LDPC decoder **810** may perform the LDPC decoding using an iterative decoding algorithm. In this case, the LDPC decoder **810** may be configured as illustrated in FIG. **9**. However, the iterative decoding algorithm is already known, and thus, the detailed configuration illustrated in FIG. **9** is only one example.

As illustrated in FIG. **9**, a decoding apparatus **900** includes an input processor **911**, a memory **912**, a variable node operator **913**, a controller **914**, a check node operator **915** and an output processor **916**.

The input processor **911** stores an input value. In detail, the input processor **911** may store the LLR values of signals received through a wireless channel.

The controller **914** determines the number of values input to the variable node operator **913**, an address value in the memory **912**, the number of values input to the check node operator **915**, an address value in the memory **912**, and the like, based on a size (that is, a codeword length) of a block of the signal received through the wireless channel and a parity-check matrix corresponding to the code rate.

According to the exemplary embodiment, indices of a row at which one (1) is positioned in a zero (0)-th column of an i-th column group may perform the decoding based on the parity-check matrices defined by above Tables 9 to 18.

The memory **912** stores the input data and output data of the variable node operator **913** and the check node operator **915**.

The variable node operator **913** receives data from the memory **912** based on information on the address of the input data and information on the number of input data which are received from the controller **914** to perform a variable node operation. Next, the variable node operator **913** stores variable node operation results in the memory **912** based on information on the address of the output data and information on the number of output data which are received from the controller **914**. Further, the variable node operator **913** inputs the variable node operation results to the output processor **916** based on data which are received from the input processor **911** and the memory **912**. Here, the variable node operation is already described with reference to FIG. **5**.

The check node operator **915** receives data from the memory **912** based on the information on the address of the input data and the information on the number of input data which are received from the controller **914** to perform a variable node operation. Next, the check node operator **915** stores variable node operation results in the memory **912** based on information on the address of the output data and information on the number of output data which are received from the controller **914**. Here, the check node operation is already described with reference to FIG. **5**.

The output processor **916** performs a soft decision on whether information word bits of the codeword at the transmitting side are zero (0) or one (1) based on data received from the variable node operator **913** and then outputs soft-decision results, such that an output value of the output processor **916** may be a value which is finally decoded. In this case, in FIG. **5**, the soft decision may be performed based on a value obtained by adding all message values (an initial message value and the other message values input from the check node) input to one variable node.

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The decoding apparatus **800** may further include a memory (not illustrated) in which information on the code rate, the codeword length, and the parity-check matrix of the LDPC code is previously stored, and the LDPC decoder **810** may perform the LDPC encoding using this information. However, this is only an example, and thus, corresponding information may be provided from the transmitting side. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (a) present in every M-th column.

FIG. **10** is a block diagram illustrating a configuration of a receiving apparatus according to an exemplary embodiment. As illustrated in FIG. **10**, the receiving apparatus **1000** includes a demodulator **1010**, a deinterleaver **1020**, an LDPC decoder **1030** and a BCH decoder **1040**.

The demodulator **1010** receives and demodulates a signal transmitted from the transmitting apparatus (for example, **700** of FIG. **7**). In detail, the demodulator **1010** may demodulate the received signal to generate a value corresponding to an LDPC codeword and output a generated value to the deinterleaver **1020**.

In this case, the value corresponding to the LDPC codeword may be represented by a channel value for the received signal. Here, a method for determining the channel value may be various and may be a method for determining an LLR value as one example.

The deinterleaver **1020** may perform deinterleaving on output values of the demodulator **1010** and output deinterleaved output values to the LDPC decoder **1030**.

In detail, the deinterleaver **1020** is a component corresponding to the interleaver **730** of the transmitting apparatus **700** and may perform an operation corresponding to the interleaver **730**. That is, the deinterleaver **1020** may inversely apply the interleaving scheme applied to the interleaver **730** to deinterleave LLR values output from the demodulator **1010**.

However, in some cases, when the interleaver **730** is omitted in the transmitting apparatus **700**, the deinterleaver **1020** may be omitted.

The LDPC decoder **1030** may use output values of the deinterleaver **1020** to perform the LDPC decoding and output LDPC decoded bits to the BCH decoder **1040**. Here, the LDPC decoded bits may be a BCH codeword.

In detail, the LDPC decoder **1030** is a component corresponding to the LDPC encoder **720** of the transmitting apparatus **700**, and may perform the LDPC decoding based on a parity-check matrix. The LDPC decoder **1030** of FIG. **10** may be implemented as the LDPC decoder **810** described with reference to FIG. **8**. That is, the LDPC decoder **1030** may perform the LDPC decoding using the parity-check matrix in which an information word sub-matrix is defined by above Tables 9 to 18 depending on a code rate and the parity sub-matrix has a dual diagonal structure.

The BCH decoder **1040** may perform the BCH decoding on values output from the LDPC decoder **1030**.

In detail, the BCH decoder **1040** is a component corresponding to the BCH encoder **710** of the transmitting apparatus **700** and may perform the BCH decoding on a BCH codeword output from the LDPC decoder **1030** to generate the bits transmitted from the transmitting apparatus **700**. However, in some cases, when the BCH encoder **710** is omitted in the transmitting apparatus **700**, the BCH decoder **1040** may be omitted.

The receiving apparatus **1000** may pre-store various parameters which are used for decoding and interleaving.

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Here, the parameters used for decoding may be information on a code rate and a codeword length of the BCH code and information on a code rate, a codeword length and a parity-check matrix of the LDPC code. Further, the parameters used for deinterleaving may be information on a deinterleaving rule. Here, the information on the parity-check matrix may comprise information about an information word sub-matrix which is formed of a plurality of column groups each including M columns and a table showing positions of value one (1) present in every M-th column.

In this case, each component configuring the receiving apparatus **1000** may be operated using the parameters.

Meanwhile, although not illustrated, in some cases, the receiving apparatus **1000** may further include a controller (not illustrated) for controlling an operation of the receiving apparatus **1000**.

In this case, the controller (not illustrated) may provide the information on the code rate and the codeword length of the BCH codes to the BCH decoder **1040** and provide the information on the code rate, the codeword length, the parity-check matrix of the LDPC codes to the LDPC decoder **1030**. Further, the controller (not illustrated) may also provide the information on the interleaving scheme to the deinterleaver **1020**. Here, the information on the parity-check matrix may comprise the information regarding information word sub-matrix which is formed of the plurality of column groups each including M columns and a table showing positions of value one (a) present in every M-th column.

FIGS. **11** and **12** are diagrams for describing performances of LDPC codes according to an exemplary embodiment.

FIG. **11** is a graph illustrating a BER performance of the LDPC codes according to an exemplary embodiment. In detail, each curve represents a BER performance to Es/No when the LDPC encoding is performed based on above Tables 9 to 18.

FIG. **12** is a graph illustrating an FER performance of the LDPC codes according to an exemplary embodiment. In detail, each curve represents an FER performance to Es/No when the LDPC encoding is performed based on above Tables 9 to 18.

As described above, when the LDPC encoding is performed based on the parity-check matrix defined according to the above exemplary embodiments, it may be appreciated that the BER/FER performance is improved.

FIG. **13** is a flow chart for describing an encoding method according to an exemplary embodiment. In detail, FIG. **13** is a diagram for describing an encoding method of an encoding apparatus for performing low density parity check (LDPC) encoding.

First, the LDPC encoding is performed on input bits based on a parity-check matrix to generate an LDPC codeword (**S1310**). In this case, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

Meanwhile, the parity-check matrix may have the same form as that of the parity-check matrix **300** illustrated in FIG. **3**.

In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix described above.

Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the

information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

In this case, in S1310, the LDPC encoding may be performed using parity-check matrices which are differently defined depending on a code rate.

For example, the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15 and the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and the LDPC encoding may be performed using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

Meanwhile, a detailed method for performing the LDPC encoding is already described, and thus, duplicate descriptions thereof will be omitted.

FIG. 14 is a flow chart for describing a decoding method according to an exemplary embodiment. In detail, FIG. 14 is a diagram for describing the decoding method of a decoding apparatus for performing low density parity check (LDPC) decoding.

First, the LDPC decoding is performed on an LDPC codeword based on a parity-check matrix (S1410). Here, the LDPC codeword may be formed of 64800 bits. That is, the LDPC codeword length may be 64800.

For example, the LDPC decoding may be performed by passing LLR values corresponding to the LDPC codeword bits through an iterative decoding algorithm to generate information word bits.

Here, the LLR values may be represented by channel values corresponding to the LDPC codeword bits by various methods.

For example, the LLR values may be represented by values obtained by taking a log on a ratio of a probability that bits transmitted through a channel at the transmitting side are zero (0) and a probability that bits are one (1). Further, the LLR values may be bit values determined by a soft decision and may also be a representative value determined depending on a section to which the probability that the bits transmitted from the transmitting side are zero (0) or one (1) belongs.

In this case, the transmitting side may use the LDPC encoder 610 as illustrated in FIG. 6 to generate and transmit the LDPC codeword.

Meanwhile, the parity-check matrix may have the same form as that of the parity-check matrix 300 illustrated in FIG. 3.

In detail, the parity-check matrix includes the information word sub-matrix and the parity sub-matrix as described above.

Here, the information word sub-matrix is formed of a plurality of column groups each including M columns and may be defined by a table indicating positions of value one (1) present in every M-th column. Here, M, which is an interval at which patterns of columns are repeated in the information word sub-matrix, may be 360. In addition, the parity sub-matrix may have a dual diagonal structure.

In this case, in S1410, the LDPC decoding may be performed using parity-check matrices which are differently defined depending on a code rate R.

For example, the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 9, 13 and 17 when the code rate is 6/15 and the LDPC decoding may be performed using parity-check matrices

defined by a table such as above Tables 10 and 14 when the code rate is 8/15. Further, the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 11, 15 and 18 when the code rate is 10/15 and the LDPC decoding may be performed using parity-check matrices defined by a table such as above Tables 12 and 16 when the code rate is 12/15.

Meanwhile, the detailed method for performing the LDPC decoding is already described, and thus, duplicate descriptions thereof will be omitted.

A non-transitory computer readable medium in which programs sequentially performing the encoding method and the decoding method according to the above exemplary embodiments are stored may be provided.

The non-transitory computer readable medium is not a medium such as a register, a cache, and a memory which may store data for a short period of time but a medium which may semi-permanently store data and read by equipment. In detail, various applications or programs as described above may be stored and provided in the non-transitory computer readable medium such as a compact disc (CD), a digital versatile disc (DVD), a hard disk, a Blu-ray disk, a universal serial bus (USB), a memory card, and a read-only memory (ROM).

Further, in the foregoing block diagram illustrating the encoding apparatus, the decoding apparatus, the transmitting apparatus, and the receiving apparatus, a bus is not illustrated, but communication between each component in the encoding apparatus, the decoding apparatus, the transmitting apparatus, and the receiving apparatus may be made through the bus.

Components, elements or units represented by a block as illustrated in FIGS. 6-10 may be embodied as the various numbers of hardware, software and/or firmware structures that execute respective functions described above, according to exemplary embodiments. For example, these components, elements or units may use a direct circuit structure, such as a memory, processing, logic, a look-up table, etc. that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. These components, elements or units may be specifically embodied by a module, a program, or a part of code, which contains one or more executable instructions for performing specified logic functions. Also, at least one of the above components, elements or units may further include a processor such as a central processing unit (CPU) that performs the respective functions, a microprocessor, or the like.

As described above, according to the exemplary embodiments, the LDPC encoding and decoding performance may be improved.

Hereinabove, although various exemplary embodiments of the inventive concept are illustrated and described, the inventive concept is not limited to the aforementioned exemplary embodiment and it is apparent that various modifications can be made to those skilled in the art without departing from the spirit of the inventive concept described in the appended claims and the modified embodiments are not to be individually understood from the technical spirit and prospects of the inventive concept.

What is claimed:

1. A transmitting apparatus comprising:

- an encoder configured to encode input bits to generate parity bits using a matrix;
- an interleaver configured to interleave a codeword comprising the input bits and the parity bits; and
- a mapper configured to map bits of the interleaved codeword to constellation points,

wherein the matrix is obtained based on parity indices of a low density parity check (LDPC) code, a code rate of the LDPC code being 6/15 and a code length of the LDPC code being 64800 bits, and

wherein the parity indices are represented by a table below:

1606	3402	4961	6751	7132	11516	12300	12482	12592	13342	13764
14123	21576	23946								
24533	25376	25667	26836	31799	34173	35462	36153	36740	37085	
37152	37468	37658								
4621	5007	6910	8732	9757	11508	13099	15513	16335	18052	19512
21319	23663	25628								
27208	31333	32219	33003	33239	33447	36200	36473	36938	37201	
37283	37495	38642								
16	1094	2020	3080	4194	5098	5631	6877	7889	8237	9804
11017	11366	13136								
13354	15379	18934	20199	24522	26172	28666	30386	32714	36390	
37015	37162									
700	897	1708	6017	6490	7372	7825	9546	10398	16605	18561
21625	22137	23693								
24340	24966	25015	26995	28586	28895	29687	33938	34520	34858	
37056	38297									
159	2010	2573	3617	4452	4958	5556	5832	6481	8227	9924
14954	15594	16623								
18065	19249	22394	22677	23408	23731	24076	24776	27007	28222	
30343	38371									
3118	3545	4768	4992	5227	6732	8170	9397	10522	11508	15536
20218	21921	28599								
29445	29758	29968	31014	32027	33685	34378	35867	36323	36728	
36870	38335	38623								
1264	4254	6936	9165	9486	9950	10861	11653	13697	13961	15164
15665	18444	19470								
20313	21189	24371	26431	26999	28086	28251	29261	31981	34015	
35850	36129	37186								
111	1307	1628	2041	2524	5358	7988	8191	10322	11905	12919
15515	15711	17061								
19024	21195	22902	23727	24401	24608	25111	25228	27338	35398	
37794	38196									
961	3035	7174	7948	13355	13607	14971	18189	18339	18665	18875
19142	20615	21136								
21309	21758	23366	24745	25849	25982	27583	30006	31118	32106	
36469	36583	37920								
2990	3549	4273	4808	5707	6021	6509	7456	8240	10044	12262
13085	14750	15680								
16049	21587	23997	25803	28343	28693	34393	34860	35490	36021	
37737	38296									
955	4323	5145	6885	8123	9730	11840	12216	19194	20313	23056
24248	24830	25268								
26617	26801	28557	29753	30745	31450	31973	32839	33025	33296	
35710	37366	37509								
264	605	4181	4483	5156	7238	8863	10939	11251	12964	16254
17511	20017	22395	22818							
23261	23422	24064	26329	27723	28186	30434	31956	33971	34372	
36764	38123									
520	2562	2794	3528	3860	4402	5676	6963	8655	9018	9783
11933	16336	17193	17320							
19035	20606	23579	23769	24123	24966	27866	32457	34011	34499	
36620	37526									
10106	10637	10906	34242							
1856	15100	19378	21848							
943	11191	27806	29411							
4575	6359	13629	19383							
4476	4953	18782	24313							
5441	6381	21840	35943							
9638	9763	12546	30120							
9587	10626	11047	25700							
4088	15298	28768	35047							
2332	6363	8782	28863							
4625	4933	28298	30289							
3541	4918	18257	31746							
1221	25233	26757	34892							
8150	16677	27934	30021							
8500	25016	33043	38070							
7374	10207	16189	35811							
611	18480	20064	38261							
25416	27352	36089	38469							
1667	17614	25839	32776							
4118	12481	21912	37945							

-continued

5573	13222	23619	31271	
18271	26251	27182	30587	
14690	26430	26799	34355	
13688	16040	20716	34558	
2740	14957	23436	32540	
3491	14365	14681	36858	
4796	6238	25203	27854	
1731	12816	17344	26025	
19182	21662	23742	27872	
6502	13641	17509	34713	
12246	12372	16746	27452	
1589	21528	30621	34003	
12328	20515	30651	31432	
3415	22656	23427	36395	
632	5209	25958	31085	
619	3690	19648	37778	
9528	13581	26965	36447	
2147	26249	26968	28776	
15698	18209	30683		
1132	19888	34111		
4608	25513	38874		
475	1729	34100		
7348	32277	38587		
182	16473	33082		
3865	9678	21265		
4447	20151	27618		
6335	14371	38711		
704	9695	28858		
4856	9757	30546		
1993	19361	30732		
756	28000	29138		
3821	24076	31813		
4611	12326	32291		
7628	21515	34995		
1246	13294	30068		
6466	33233	35865		
14484	23274	38150		
21269	36411	37450		
23129	26195	37653		

2. The transmitting apparatus of claim 1, wherein the matrix has a cyclic structure.

3. The transmitting apparatus of claim 1, wherein a predetermined number which is differently set according to the code rate and the code length is 108 when the code rate is 6/15 and the code length is 64800 bits.

4. The transmitting apparatus of claim 1, wherein the parity bits are generated by initiating parity bits, accumulating the input bits to the initiated parity bits based on the matrix, and performing a bit operation based on an accumulated parity bit of a parity index x and an accumulated parity bit of a parity index x-1 to output a parity bit of the parity index x,

where x is a value among values which are greater than 0 and are less than a number of parity bits.

5. The transmitting apparatus of claim 4, wherein the matrix is represented by $q(i,j,l)$ generated based on a following relationship:

$$q(i,j,l)=q(i,j,0)+Q_{ldpc} \times l \pmod{M_{inner}} \text{ for } 0 \leq l < 360$$

where Q_{ldpc} is a coding parameter determined by the code rate of the LDPC code and the code length of the LDPC code, M_{inner} is the number of parity bits, $q(i,j,0)$ is a parity index represented by a j-th entry in an i-th row of the table, i indicates an index of a row in the table, and j indicates an index of an entry in the row.

6. The transmitting apparatus of claim 5, wherein the input bits are accumulated to the initiated parity bits based on a following relationship,

$$P_{q(i,w(i),l)}=P_{q(i,w(i),l)}+s_k$$

where s_k indicates k-th input bit from among the input bits, $p_{q(i,w(i),l)}$ indicates $q(i,w(i),l)$ -th parity bit from

among the initiated parity bits, k is a value among values which are greater than or equal to 0 and are less than a number of the input bits, $i = \lfloor k/360 \rfloor$, $l = k \bmod 360$, and $w(i) = j$.

7. A receiving apparatus comprising:
a demodulator configured to demodulate a signal received from a transmitting apparatus to generate values;
a deinterleaver configured to deinterleave the values; and

a decoder configured to decode the deinterleaved values based on a matrix of a low density parity check (LDPC) code, a code rate of the LDPC code being 6/15 and a code length of the LDPC code being 64800 bits, wherein the matrix is obtained based on parity indices, and wherein the parity indices are represented by a table below:

1606 3402 4961 6751 7132 11516 12300 12482 12592 13342 13764 14123 21576 23946 24533
25376 25667 26836 31799 34173 35462 36153 36740 37085 37152 37468 37658
4621 5007 6910 8732 9757 11508 13099 15513 16335 18052 19512 21319 23663 25628 27208
31333 32219 33003 33239 33447 36200 36473 36938 37201 37283 37495 38642
16 1094 2020 3080 4194 5098 5631 6877 7889 8237 9804 10067 11017 11366 13136 13354
15379 18934 20199 24522 26172 28666 30386 32714 36390 37015 37162
700 897 1708 6017 6490 7372 7825 9546 10398 16605 18561 18745 21625 22137 23693 24340
24966 25015 26995 28586 28895 29687 33938 34520 34858 37056 38297
29758 29968 31014 32027 33685 34378 35867 36323 36728 36870 38335 38623
1264 4254 6936 9165 9486 9950 10861 11653 13697 13961 15164 15665 18444 19470 20313
21189 24371 26431 26999 28086 28251 29261 31981 34015 35850 36129 37186
111 1307 1628 2041 2524 5358 7988 8191 10322 11905 12919 14127 15515 15711 17061 19024
21195 22902 23727 24401 24608 25111 25228 27338 35398 37794 38196
961 3035 7174 7948 13355 13607 14971 18189 18339 18665 18875 19142 20615 21136 21309
21758 23366 24745 25849 25982 27583 30006 31118 32106 36469 36583 37920
2990 3549 4273 4808 5707 6021 6509 7456 8240 10044 12262 12660 13085 14750 15680 16049
21587 23997 25803 28343 28693 34393 34860 35490 36021 37737 38296
955 4323 5145 6885 8123 9730 11840 12216 19194 20313 23056 24248 24830 25268 26617
26801 28557 29753 30745 31450 31973 32839 33025 33296 35710 37366 37509
264 605 4181 4483 5156 7238 8863 10939 11251 12964 16254 17511 20017 22395 22818 23261
23422 24064 26329 27723 28186 30434 31956 33971 34372 36764 38123
520 2562 2794 3528 3860 4402 5676 6963 8655 9018 9783 11933 16336 17193 17320 19035
20606 23579 23769 24123 24966 27866 32457 34011 34499 36620 37526
10106 10637 10906 34242
1856 15100 19378 21848
943 11191 27806 29411
4575 6359 13629 19383
4476 4953 18782 24313
5441 6381 21840 35943
9638 9763 12546 30120
9587 10626 11047 25700
4088 15298 28768 35047
2332 6363 8782 28863
4625 4933 28298 30289
3541 4918 18257 31746
1221 25233 26757 34892
8150 16677 27934 30021
8500 25016 33043 38070
7374 10207 16189 35811
611 18480 20064 38261
25416 27352 36089 38469
1667 17614 25839 32776
4118 12481 21912 37945
5573 13222 23619 31271
18271 26251 27182 30587
14690 26430 26799 34355
13688 16040 20716 34558
2740 14957 23436 32540
3491 14365 14681 36858
4796 6238 25203 27854
1731 12816 17344 26025
19182 21662 23742 27872
6502 13641 17509 34713
12246 12372 16746 27452
1589 21528 30621 34003
12328 20515 30651 31432
3415 22656 23427 36395
632 5209 25958 31085
619 3690 19648 37778
9528 13581 26965 36447
2147 26249 26968 28776
15698 18209 30683
1132 19888 34111
4608 25513 38874
475 1729 34100
7348 32277 38587
182 16473 33082

3865 9678 21265
4447 20151 27618
6335 14371 38711
704 9695 28858
4856 9757 30546
1993 19361 30732
756 28000 29138
3821 24076 31813
4611 12326 32291
7628 21515 34995
1246 13294 30068
6466 33233 35865
14484 23274 38150
21269 36411 37450
23129 26195 37653.

8. The receiving apparatus of claim 7, wherein the matrix has a cyclic structure.

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